MITx MicroMaster in Statistics and Data Science

Tommaso Facchin

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Abstract

These are my personal study notes for the MITx MicroMaster in Statistics and Data Science. The material follows the main courses of the program. The text was drafted with the help of an AI assistant and then adapted and edited for my own use. The goal is to have a concise and easy-to-read reference, useful for quick review, that follows the structure of the original courses.
The sections are organized hierarchically as *course.unit.x*.
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1 Probability - The Science of Uncertainty and Data (6.431x)

1.1 Probability models and axioms

A probabilistic model describes the behavior of a random experiment, such as flipping a coin, rolling dice, or throwing a dart. Building such a model has two steps: first, we list all possible outcomes; second, we assign probabilities to them. In this section, we focus on the first step — describing the possible outcomes clearly and systematically.

1.1.1 Sample space

All possible outcomes of an experiment form a set, denoted by Ω, called the sample space. Each element of Ω represents one possible result. The outcomes must be mutually exclusive (only one can happen at a time) and collectively exhaustive (some outcome from Ω must always happen).
How detailed our sample space is depends on what we care about in the experiment. It should include all relevant distinctions but ignore unimportant ones. Choosing a good sample space is part of the modeling process.

If we flip a coin, a simple sample space is

$$\Omega = \{\text{heads}, \text{tails}\}.$$

This covers all outcomes and they don't overlap.
If we also record whether it's raining, we could define a larger sample space,

$$\Omega' = \{\text{heads and rain}, \text{heads and no rain}, \text{tails and rain}, \text{tails and no rain}\}.$$

This is valid too, but since weather usually doesn't affect a coin toss, it's an unnecessary detail. We include such extra factors only when they matter for the problem.

1.1.2 Discrete and continuous sample spaces

Sample spaces can be discrete or continuous.
For example, rolling a four-sided die twice and recording the two results gives

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4\}\},$$

with 16 possible ordered pairs like $(1, 1)$ or $(3, 4)$. When experiments have several stages, tree diagrams can help visualize the structure of outcomes.
Other sample spaces are continuous. For instance, when throwing a dart at a unit square, each landing point is a pair of coordinates (x, y):

$$\Omega = \{(x, y) : 0 \leq x \leq 1,\ 0 \leq y \leq 1\}.$$

Here, every point inside the square is a possible outcome.

1.1.3 Events and probability laws

Once we have a sample space, we can talk about events. An event A is any subset of Ω. After an experiment, if the observed outcome ω belongs to A, we say the event occurred.
In discrete cases, we can assign probabilities to single outcomes. In continuous cases, each exact point has probability 0, so we assign probabilities to regions instead (for example, "the dart lands in the upper half of the square"). The probability law gives each event A a value $P(A)$, showing how likely that event is.

1.1.4 Probability axioms and basic set notation

Probabilities follow three basic rules, called axioms:

- **Nonnegativity**: For any event A, $P(A) \geq 0$.
- **Normalization**: $P(\Omega) = 1$, since something in the sample space must occur.
- **Additivity**: If two events A and B are disjoint (that is, $A \cap B = \emptyset$), then

$$P(A \cup B) = P(A) + P(B).$$

You can think of probability as a kind of "mass" spread over Ω. If two events don't overlap, the total mass on their union is just the sum of their individual masses.

For two events A and B:

- The intersection $A \cap B$ is the set of outcomes that belong to both A and B.
- The union $A \cup B$ is the set of outcomes that belong to either A or B (or both).
- Two events are disjoint if $A \cap B = \emptyset$, meaning they have no outcomes in common.

1.1.5 Simple properties derived from the axioms

The three axioms look simple, but they already imply many useful and intuitive properties. For example, we can prove that probabilities are always at most 1, and that the probability of the empty set is 0, instead of assuming these as extra axioms.
Consider an event A and its complement A^c, which contains all outcomes in Ω that are not in A. The set A and its complement A^c are disjoint, and their union is the entire sample space:

$$A \cup A^c = \Omega, \qquad A \cap A^c = \emptyset.$$

Using normalization and additivity, we get

$$P(\Omega) = P(A \cup A^c) = P(A) + P(A^c) = 1.$$

From this relation we also obtain

$$P(A) = 1 - P(A^c).$$

Because probabilities are nonnegative, $P(A^c) \geq 0$, so $P(A) \leq 1$. This shows that probabilities are always bounded above by 1, without adding a separate axiom.
If we apply the same relation to the special case where $A = \Omega$, then the complement A^c is the empty set $\emptyset$. We already know that $P(\Omega) = 1$, so

$$1 = P(\Omega) = P(\Omega) + P(\emptyset),$$

which forces $P(\emptyset) = 0$. This matches the idea that the empty set is "impossible," so its probability should be zero.
Finally, the additivity axiom for two disjoint events extends to any finite collection of disjoint events. If $A_1, A_2, \ldots, A_k$ are pairwise disjoint, then

$$P\Big(\bigcup_{i=1}^{k} A_i\Big) = \sum_{i=1}^{k} P(A_i).$$

A common special case is when we have a finite subset of the sample space consisting of points $s_1, \ldots, s_k$. We can view this set as the union of the singletons $\{s_1\}, \ldots, \{s_k\}$, which are disjoint. Then the probability of the finite set is just the sum of the probabilities of its individual elements. In practice we often write $P(s_i)$ instead of $P(\{s_i\})$ and talk informally about "the probability of an outcome s_i" even though probabilities are formally defined on sets.

The axioms also give us some useful monotonicity and union properties. Suppose we have two events A and B with $A \subseteq B$. Intuitively, B is "larger," so its probability should be at least as big as that of A. Formally, we can write

$$B = A \cup \big(B \cap A^c\big),$$

where A and $B \cap A^c$ are disjoint. By additivity,

$$P(B) = P(A) + P\big(B \cap A^c\big) \geq P(A),$$

since probabilities are nonnegative.
We can also express the probability of a union when the sets are not disjoint. If A and B overlap, then the union $A \cup B$ can be decomposed into three disjoint parts: the part of A outside B, the intersection $A \cap B$, and the part of B outside A. If we call their probabilities a, b, and c, then

$$P(A \cup B) = a + b + c.$$

On the other hand,

$$P(A) = a + b, \qquad P(B) = b + c, \qquad P(A \cap B) = b.$$

Putting these together gives the standard identity

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

A direct corollary is the union bound. Since $P(A \cap B) \geq 0$, we get

$$P(A \cup B) \leq P(A) + P(B),$$

which is useful when we only need an upper bound on a probability.
The same idea extends to three sets. For events A, B, and C, we can rewrite their union as a union of three disjoint pieces: A itself, the part of B outside A, and the part of C outside both A and B. Using additivity on these disjoint parts gives a formula for $P(A \cup B \cup C)$ in terms of simpler probabilities. More generally, we can keep breaking unions into disjoint pieces to get expressions for probabilities of unions of several events.

1.1.6 A discrete example and the discrete uniform law

Consider again two rolls of a tetrahedral die. The sample space has 16 outcomes, each an ordered pair (X, Y), where X is the first roll and Y is the second. Assume all 16 outcomes are equally likely, so each has probability $1/16$.
As a first example, look at the event $\{X = 1\}$: the first roll is 1. In the grid of all (X, Y) pairs, this event contains the four outcomes where the first coordinate is 1 and the second is 1, 2, 3, or 4. Since each outcome has probability $1/16$, we get

$$P(X = 1) = 4 \cdot \frac{1}{16} = \frac{1}{4}.$$

Now define Z to be the smaller of the two numbers in the two rolls. For example, if $X = 2$ and $Y = 3$, then $Z = 2$. For $P(Z = 4)$, this happens only if both X and Y are 4, so there is exactly one outcome in the event and

$$P(Z = 4) = \frac{1}{16}.$$

For $\{Z = 2\}$, we need one roll equal to 2 and the other at least 2. This event contains $(2, 2)$, $(2, 3)$, $(2, 4)$, $(3, 2)$, and $(4, 2)$, for a total of five outcomes. Thus

$$P(Z = 2) = 5 \cdot \frac{1}{16}.$$

This is an instance of the discrete uniform law. If the sample space Ω is finite with n elements and all are equally likely, then, since $P(\Omega) = 1$, we must have

$$P(\{\omega\}) = \frac{1}{n}$$

for each outcome ω. If an event A has exactly k elements, then

$$P(A) = k \cdot \frac{1}{n} = \frac{k}{n}.$$

So in a discrete uniform model, computing probabilities amounts to counting: we count all possible outcomes n, and the favorable ones k, then take the ratio k/n.

1.1.7 A continuous example

Now consider a probability calculation in a continuous sample space. The experiment is again throwing a dart at the unit square, and the outcome is the point (x, y) where the dart lands. The sample space is the unit square, and we assume a uniform law: the probability of any region in the square is equal to its area. This is natural if we think the dart is equally likely to land anywhere.
Take the event that the sum of the coordinates is at most $1/2$:

$$A = \{(x, y) : x + y \leq 1/2\}.$$

In the unit square this is a right triangle under the line $x + y = 1/2$. Since probability equals area,

$$P(A) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

As another example, consider the event that the dart hits exactly the point $(0.5, 0.3)$. This event is a single point with area 0, so its probability is 0. The same holds for any single point in the unit square: in a continuous model, individual points have probability 0, while regions with positive area can have positive probability.
This example fits a general four-step recipe for probability problems. Starting from a word description of an experiment, we:

1. Specify the sample space.
2. Choose a probability law (a model for how likely different outcomes are).
3. Describe the event of interest precisely, ideally also with a picture.
4. Use the probability law and the event description to compute the desired probability.

Sometimes the law is given explicitly for all events; in other cases we only know it for some basic events and need extra work to obtain the probability we want.

1.1.8 Countable additivity and infinite sample spaces

We now look at a discrete but infinite sample space. Imagine an experiment whose outcome is a positive integer, for example the number of coin tosses until the first heads appears. The sample space is $\{1, 2, 3, \dots\}$. Suppose we are told that

$$P(\{n\}) = \frac{1}{2^n}, \quad n = 1, 2, 3, \dots$$

First we check that this is a valid probability law by summing over all n:

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = \frac{1}{2} \cdot \frac{1}{1 - 1/2} = 1.$$

Now consider the event that the outcome is even, the infinite set $\{2, 4, 6, \dots\}$. We can write it as a union of singletons:

$$\{2, 4, 6, \dots\} = \{2\} \cup \{4\} \cup \{6\} \cup \dots$$

These sets are disjoint, so we would like to write

$$P(\text{even}) = P(\{2\}) + P(\{4\}) + P(\{6\}) + \dots$$

and evaluate the resulting infinite series. To make this step valid in general, we extend the additivity axiom from finite unions to countably infinite unions.
The extra axiom, called countable additivity, says: if $A_1, A_2, A_3, \dots$ is a sequence of pairwise disjoint events, then

$$P\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{n=1}^{\infty} P(A_n).$$

The word "sequence" matters: the events can be listed in order $A_1, A_2, \dots$, so we can form a countable sum. This axiom lets us compute probabilities of many events in infinite sample spaces by summing over simpler disjoint pieces.
It is also important to see what countable additivity does not say. In the continuous unit square model, the square is the union of singletons (one per point), and each singleton has probability 0. If we tried to add their probabilities, we would get a sum of zeros, but the probability of the whole square should be 1. The catch is that the unit square is uncountable: its points cannot be arranged as a sequence $A_1, A_2, \dots$, so the axiom does not apply. In short, some infinite sets are countable (like the integers) and others are uncountable (like the real line or the unit square), and countable additivity only covers the countable case.
There are deeper measure-theoretic issues in the background, especially when we try to define probability on all subsets of a continuous space. For this course, the key point is that standard constructions, like defining probability on the unit square via area, can be made rigorous and do satisfy countable additivity for all "nice" sets we will use. The strange sets where problems arise are mostly of theoretical interest and do not appear in typical applications.

1.1.9 Interpretations and uses of probability

To close, it is useful to think about what probability really means. In the narrow sense, probability theory is a branch of mathematics: we start from axioms, study models that satisfy them, and prove theorems. This view does not fix any particular real-world interpretation.
One common interpretation is in terms of long-run frequencies. For example, if we have a fair coin and toss it many times, the fraction of heads is expected to approach 1/2. In this view, the probability of an event is the limiting frequency with which it occurs in a long sequence of repeated trials. This fits well for repeatable experiments like coin tosses.
But not all probability statements are like that. A sentence such as "the current president of my country will be reelected with probability 0.7" does not describe an experiment we can repeat infinitely many times. In such cases it is more natural to see probabilities as measuring degrees of belief: they summarize how strongly we believe an event will happen. Thinking in terms of bets, a probability expresses how much we are willing to stake on an outcome.
This raises the worry that probabilities are just subjective. The answer is that probability theory at least gives consistent rules for reasoning under uncertainty. If our probability model is a reasonable description of the situation, those rules help us make predictions and decisions in a systematic way.
Whether our decisions are actually good depends on whether we chose a good model. Building such models from data is the task of statistics and statistical inference. The real world generates data; statistics uses these data to construct probabilistic models; and probability theory then analyzes these models to produce predictions or decisions that we can apply back to the real world.

1.1.10 Sets, complement, empty set, and subsets

A set is a collection of distinct elements, usually denoted by a capital letter like S. We can specify a set by listing its elements inside braces, for example $\{a, b, c, d\}$, which is a finite set. Other sets, such as the set of all real numbers, are infinite. We write $x \in S$ if x is an element of S, and $x \notin S$ if x does not belong to S.
Sometimes it is convenient to define a set by a property instead of listing elements. For example, starting from the real numbers, we can consider all x such that $\cos(x) > 1/2$; the collection of all such x is a set. Often we also fix a larger "universe" of objects, called the universal set and denoted by Ω, and only look at sets contained in this universe.
Given a universal set Ω and a subset $S \subseteq \Omega$, the complement of S, denoted S^c, is the set of all elements in Ω that are not in S:

$$S^c = \{x \in \Omega : x \notin S\}.$$

Taking complements twice brings us back to the original set: $(S^c)^c = S$. A special set is the empty set $\emptyset$, which contains no elements. For example, the complement of Ω is $\emptyset$, since there is nothing outside the universal set.
If every element of a set S also belongs to a set T, we say that S is a subset of T and write $S \subseteq T$. This includes the case $S = T$. Intuitively, S is "contained" inside T.

1.1.11 Union, intersection, and infinite collections

Given two sets S and T, their union $S \cup T$ is the set of all elements that belong to at least one of them:

$$x \in S \cup T \quad \text{iff} \quad x \in S \text{ or } x \in T.$$

Their intersection $S \cap T$ is the set of elements that belong to both:

$$x \in S \cap T \quad \text{iff} \quad x \in S \text{ and } x \in T.$$

These notions extend to infinite families of sets. If we have sets $S_1, S_2, S_3, \ldots$, their union is

$$\bigcup_{n=1}^{\infty} S_n = \{x : x \in S_n \text{ for at least one } n\},$$

and their intersection is

$$\bigcap_{n=1}^{\infty} S_n = \{x : x \in S_n \text{ for all } n\}.$$

Informally, the union collects elements that appear in at least one of the sets, while the intersection keeps only the elements common to all of them.

1.1.12 Basic algebra of sets

Set operations satisfy some familiar algebraic properties. Union and intersection are both commutative:

$$S \cup T = T \cup S, \qquad S \cap T = T \cap S,$$

and associative:

$$(S \cup T) \cup U = S \cup (T \cup U), \qquad (S \cap T) \cap U = S \cap (T \cap U).$$

Because of associativity, we usually drop parentheses when writing unions or intersections of three or more sets.
The universal set Ω behaves like an identity: $S \cup \Omega = \Omega$ and $S \cap \Omega = S$. There are also distributive laws describing how union and intersection interact, for example

$$S \cap (T \cup U) = (S \cap T) \cup (S \cap U),$$

and a similar law with $\cup$ and $\cap$ swapped. A standard way to check such identities is to argue element-wise: assume x belongs to one side, unpack the definitions, and show that this forces x to belong to the other side as well. If each side is a subset of the other, then the two sets are equal.

1.1.13 De Morgan's laws

De Morgan's laws relate unions, intersections, and complements. For two sets S and T, the first law says that the complement of their intersection is the union of their complements:

$$(S \cap T)^c = S^c \cup T^c.$$

Intuitively, if a point is not in both S and T at the same time, then it must be missing from at least one of them, so it lies in S^c or in T^c. A formal proof rewrites the membership condition step by step: $x \notin S \cap T$ means "x is not in S or not in T," which is the same as "$x \in S^c$ or $x \in T^c$," that is, $x \in S^c \cup T^c$.
The second law is symmetric and says that the complement of a union is the intersection of the complements:

$$(S \cup T)^c = S^c \cap T^c.$$

Here the idea is that if a point is not in $S \cup T$, then it cannot be in S and cannot be in T, so it must belong to both complements. Both identities extend to families of sets $\{S_n\}$:

$$\Big(\bigcap_n S_n\Big)^c = \bigcup_n S_n^c, \qquad \Big(\bigcup_n S_n\Big)^c = \bigcap_n S_n^c.$$

These laws are very useful when we want to switch between unions and intersections, especially in probability.

1.1.14 Sequences and their limits

A sequence is a collection of elements indexed by the positive integers. We usually write it as $\{a_i\}_{i=1}^{\infty}$, or simply a_i, where each a_i belongs to some set S. Often $S = R$ and we have a sequence of real numbers, but S could also be R^n (a sequence of vectors) or some other set. Formally, a sequence is just a function f from the natural numbers to S, and the i-th term is $a_i = f(i)$.
We say that a sequence a_i converges to a real number a if, as i becomes large, the terms of the sequence get arbitrarily close to a and then stay close. Formally, $a_i \to a$ as $i \to \infty$ if for every $\varepsilon > 0$ there exists an index i_0 such that for all $i \geq i_0$,

$$|a_i - a| < \varepsilon.$$

Geometrically, if you draw a horizontal band of width 2ε around the limit a, then from some point on all the sequence terms lie inside that band. The definition may look technical, but it just encodes the idea that the sequence eventually stays as close to a as we like.
Convergence behaves well under basic operations. If $a_i \to a$ and $b_i \to b$, then the sum sequence $a_i + b_i$ converges to $a+b$, and the product sequence $a_i b_i$ converges to ab. If g is a continuous function and $a_i \to a$, then the sequence $g(a_i)$ converges to $g(a)$; for example, if $a_i \to a$, then $a_i^2 \to a^2$. These properties let us manipulate limits in a straightforward way.

1.1.15 Practical convergence criteria

In practice, it is helpful to have quick tests for convergence. One common situation is a monotone sequence. If a_i is increasing (or at least never decreases) and is bounded above, then it converges to some finite limit. If it keeps increasing without any bound, we say that it "converges to infinity" rather than to a finite number.
Another strategy is to bound the distance $|a_i - a|$ from a candidate limit a. If we can find an upper bound on $|a_i - a|$ that goes to 0 as i grows, then a_i must converge to a. A related idea is the sandwich (or squeeze) argument: if a sequence a_i is always between two other sequences b_i and c_i, and both b_i and c_i converge to the same limit a, then a_i also converges to a. These tools will appear often, sometimes implicitly, when analyzing limits in the course.

1.1.16 Infinite Series

Consider a sequence a_i and the sum:

$$\sum_{i=1}^{\infty} a_i.$$

This sum is defined as the limit of the partial sums:

$$S_n = \sum_{i=1}^{n} a_i,$$

if the limit exists and is finite as $n \to \infty$. The series converges if this limit exists; otherwise, it diverges.
For non-negative terms, partial sums are increasing and bounded, ensuring convergence (finite or infinite). For alternating signs, the limit may depend on the order of terms, so absolute convergence is preferred. If

$$\sum_{i=1}^{\infty} |a_i|$$

converges, then the series $\sum a_i$ converges as well.

A key example is the geometric series. For $|\alpha| < 1$, consider:

$$\sum_{i=0}^{\infty} \alpha^i = 1 + \alpha + \alpha^2 + \dots$$

The sum is:

$$\sum_{i=0}^{\infty} \alpha^i = \frac{1}{1-\alpha}.$$

Alternatively, using the trick $s = \sum_{i=0}^{\infty} \alpha^i$, we get:

$$s = \frac{1}{1-\alpha}, \quad |\alpha| < 1.$$

1.1.17 Series with Multiple Indices and Order of Summation

For series with two indices, like a_{ij} with $i, j = 1, 2, \dots$, we sum over a grid of points (i, j). The sum can be written as:

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} \quad \text{or} \quad \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$

Different summation orders may yield different results, but absolute convergence ensures the sum is well-defined, and the order does not matter:

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |a_{ij}|.$$

Without absolute convergence, changing the order can lead to different sums, such as 0 or 1, due to cancellations between positive and negative terms.
We can also sum over restricted sets, e.g., $\sum_{i=1}^{\infty} \sum_{j=1}^{i} a_{ij}$, and the order of summation can vary. With absolute convergence, the result is independent of the summation order.

1.1.18 Countable and Uncountable Sets

In probability theory, not all infinite sets are the same. Some are countable, others uncountable. A set is countable if its elements can be mapped one-to-one to natural numbers, i.e., they can be listed in a sequence.
Positive integers are countable, as are all integers, which can be listed as $0, 1, -1, 2, -2, 3, -3, \dots$. The same applies to pairs (i, j) of positive integers, which can be arranged in a zigzag pattern.
A less obvious example is the set of rationals in $[0, 1]$. Since every rational can be written as p/q, we can list them by varying q and p, skipping duplicates, resulting in a sequence containing all rationals in $[0, 1]$.
A set is uncountable if it cannot be listed in this way. Typical examples include intervals of real numbers, the real line, and spaces like the plane or 3D space. These sets are too large to be listed. The distinction between countable and uncountable sets is key to understanding the difference between discrete and continuous probability models.

1.1.19 Cantor's Diagonal Argument

Cantor's diagonal argument proves that the set of real numbers in $[0, 1]$ is uncountable. The idea is to assume that we can list all real numbers in this interval, each represented by an infinite decimal expansion. If we arrange them in a sequence, like:

$$r_1 = 0.a_{11}a_{12}a_{13}\dots, \quad r_2 = 0.a_{21}a_{22}a_{23}\dots, \quad r_3 = 0.a_{31}a_{32}a_{33}\dots, \dots$$

Then, by constructing a new number r whose n-th decimal place is different from the n-th decimal of r_n, we ensure that r is not on the list. This contradiction shows that the real numbers in $[0, 1]$ cannot be counted, proving they are uncountable.
Thus, Cantor's argument demonstrates that no list can exhaust all real numbers, highlighting the distinction between countable and uncountable sets.

1.2 Conditioning and independence

1.2.1 Conditional probabilities

Conditional probabilities formalize how we update a probability model once we learn that some event has occurred. Conditional probabilities are probabilities associated with a revised model that takes into account extra information about the outcome of a probabilistic experiment. The question is how to systematically revise our model when we are told that a certain event has occurred.

Consider a probability model with a sample space Ω and a probability law P. Let A and B be two events, that is, two subsets of Ω. We will be interested in the probability of A under the information that B has occurred.

To build intuition, start with a finite sample space with equally likely outcomes. Suppose that Ω has 12 possible outcomes, each with probability $1/12$. Let A be an event consisting of 5 outcomes, so that $P(A) = 5/12$, and let B be an event consisting of 6 outcomes, so that $P(B) = 6/12$. Assume that A and B overlap in exactly 2 outcomes, so that

$$P(A \cap B) = \frac{2}{12}.$$

Now suppose we are told that B has occurred, but nothing more. How should we revise the model? First, all outcomes outside B are no longer possible. In the updated model they must have probability 0. Second, inside B we still have 6 outcomes. In the original model they were equally likely, and the new information does not change their relative likelihoods, so they should remain equally likely. Since the total probability inside B must now be 1, each of the 6 outcomes in B should have probability $1/6$ in the revised model.

In this revised model the probability of A is just the fraction of the outcomes in B that also belong to A. There are 2 such outcomes, so

$$P(A \mid B) = \frac{2}{6} = \frac{1}{3}.$$

Here we have introduced the notation $P(A \mid B)$ for the probability of A given B, read as "the probability of A given B". Similarly, since B is now certain in the revised model, we have $P(B \mid B) = 1$.

The same idea works even if the outcomes of Ω are not equally likely. Suppose we only know the probabilities of certain regions of the sample space without assuming any uniformity inside them. For instance, we might still have

$$P(A) = \frac{5}{12}, \qquad P(B) = \frac{6}{12}, \qquad P(A \cap B) = \frac{2}{12},$$

but now these numbers come directly from the probability law P without referring to equally likely elementary outcomes. Inside B the subset where A also occurs, namely $A \cap B$, has total probability $P(A \cap B) = 2/12$. Relative to the total probability of B, which is $P(B) = 6/12$, this corresponds to a fraction

$$\frac{P(A \cap B)}{P(B)} = \frac{2/12}{6/12} = \frac{1}{3}.$$

It is therefore natural to define the conditional probability of A given B to be this fraction. Intuitively, once we restrict attention to B we renormalize the probability law by dividing by $P(B)$ and then ask what portion of the "mass" inside B lies in A.

This leads to the formal definition. Given two events A and B with $P(B) > 0$, the conditional probability of A given B is defined by

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

The numerator $P(A \cap B)$ is the probability that both A and B occur, while the denominator $P(B)$ is the probability that B occurs. Their ratio measures the fraction of the probability of B that also belongs to A. This definition only makes sense when $P(B) > 0$. If $P(B) = 0$, the expression $P(A \cap B)/P(B)$ is undefined, and conditional probabilities given B are not defined.

It is important to emphasize that this is a definition, not a theorem. There is no question of "proving" that

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

is correct. Rather, we adopt this formula because it captures the intuitive procedure of restricting the model to B and renormalizing the probabilities. The earlier examples simply motivate why this is the most natural way to construct a conditional probability law that reflects the information that the event B has occurred.

Conditional probabilities obey the same axioms as ordinary probabilities and therefore behave as an ordinary probability law on a new sample space.

Conditional probabilities are defined from an underlying probability model (Ω, P) and an event B with $P(B) > 0$. For any event A, the conditional probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

This definition suggests that we can think of a new probabilistic model in which the outcomes are restricted to the event B and the probabilities are renormalized. In this new model the function $A \mapsto P(A \mid B)$ should behave like a probability law. We now check that it satisfies the usual axioms.
First, conditional probabilities are nonnegative. Indeed, both $P(A \cap B)$ and $P(B)$ are nonnegative, and $P(B) > 0$ by assumption, so the ratio

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

is always greater than or equal to 0. Thus the nonnegativity axiom remains valid under conditioning.
Second, we verify that the whole sample space has probability 1 in the conditional model. Consider the conditional probability of Ω given B:

$$P(\Omega \mid B) = \frac{P(\Omega \cap B)}{P(B)}.$$

Since $B \subseteq \Omega$, the intersection $\Omega \cap B$ is just B, so the numerator is $P(B)$. Therefore

$$P(\Omega \mid B) = \frac{P(B)}{P(B)} = 1.$$

We can also view the new sample space as B itself, because outcomes outside B are considered impossible once we know that B has occurred. In that view the "entire sample space" is B, and its conditional probability is

$$P(B \mid B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

So the normalization axiom $P(\text{whole space}) = 1$ holds whether we regard Ω or B as the relevant sample space under conditioning.
Third, we check the additivity axiom. Let A and C be two disjoint events, so that $A \cap C = \emptyset$. We want to show that

$$P(A \cup C \mid B) = P(A \mid B) + P(C \mid B).$$

Using the definition of conditional probability, we have

$$P(A \cup C \mid B) = \frac{P\big((A \cup C) \cap B\big)}{P(B)}.$$

By a basic set identity,

$$(A \cup C) \cap B = (A \cap B) \cup (C \cap B).$$

Now note that A and C are disjoint, so their intersections with B are also disjoint. Indeed, any point in $(A \cap B) \cap (C \cap B)$ would belong to both A and C, which is impossible. Therefore $(A \cap B)$ and $(C \cap B)$ are disjoint sets, and by the additivity axiom for the original probability measure P,

$$P\big((A \cup C) \cap B\big) = P(A \cap B) + P(C \cap B).$$

Substituting into the expression for the conditional probability gives

$$P(A \cup C \mid B) = \frac{P(A \cap B) + P(C \cap B)}{P(B)} = \frac{P(A \cap B)}{P(B)} + \frac{P(C \cap B)}{P(B)} = P(A \mid B) + P(C \mid B).$$

This shows that conditional probabilities preserve finite additivity for disjoint events. The same reasoning extends, without essential changes, to finite unions of pairwise disjoint events and even to countable unions when countable additivity holds for the original probability measure P. Thus, conditional probabilities satisfy nonnegativity, normalization, and (finite or countable) additivity, and therefore they obey all the standard axioms of probability theory.
An important consequence is that conditional probabilities can be treated as ordinary probabilities. Any identity or theorem that we derive for ordinary probabilities continues to hold if we replace $P(\cdot)$ by $P(\cdot \mid B)$. In later sections we will repeatedly use this idea to apply general probability rules, such as the product rule or Bayes' rule, in conditional form.

EXAMPLE: RADAR DETECTION PROBLEM

Conditional probabilities are useful both for updating a model when new information arrives and for building multi-stage probabilistic models.

Consider a simple radar detection problem. Either there is an airplane in the sky or there is not. Let A be the event that an airplane is present and A^c the event that there is no airplane. Suppose prior information tells us that

$$P(A) = 0.05, \qquad P(A^c) = 0.95.$$

A radar looks at the sky and either reports a detection or not. Let B be the event that the radar reports a detection and B^c the event that it does not. The performance of the radar is summarized by conditional probabilities that describe how it behaves in the two possible worlds A and A^c.

If an airplane is present, the radar should detect it most of the time, but it can also miss it. For example, we might have

$$P(B \mid A) = 0.99, \qquad P(B^c \mid A) = 0.01,$$

so that conditionally on A the radar detects the plane with probability 99% and misses it with probability 1%. If no airplane is present, the radar can still produce a false alarm or remain silent. For instance,

$$P(B \mid A^c) = 0.10, \qquad P(B^c \mid A^c) = 0.90,$$

so there is a 10% chance of a false alarm and a 90% chance that the radar correctly reports no detection when nothing is flying.

These four conditional probabilities, together with the prior probabilities of A and A^c, define a complete probabilistic model. We can represent it with a tree diagram: first choose between A and A^c, then from each of these nodes choose between B and B^c using the corresponding conditional probabilities. The probability of each "leaf" of the tree is obtained by multiplying along the path that leads to it. This is an instance of the product rule

$$P(A \cap B) = P(A)\, P(B \mid A),$$

which follows directly from the definition of conditional probability.

As a first example, let us compute the probability that there is an airplane and that the radar detects it, that is, $P(A \cap B)$. Using the product rule,

$$P(A \cap B) = P(A)\, P(B \mid A) = 0.05 \times 0.99 = 0.0495.$$

In the tree picture, this is the probability of the path where A occurs at the first stage and B occurs at the second stage.

Next, consider the total probability that the radar reports a detection, $P(B)$. There are two disjoint ways in which this can happen: either there is an airplane and it is detected, or there is no airplane and the radar produces a false alarm. In symbols,

$$B = (A \cap B) \cup (A^c \cap B),$$

and the two events on the right-hand side are disjoint. Therefore

$$P(B) = P(A \cap B) + P(A^c \cap B).$$

Using the product rule for each term, we obtain

$$P(B) = P(A)\, P(B \mid A) + P(A^c)\, P(B \mid A^c) = 0.05 \times 0.99 + 0.95 \times 0.10.$$

Numerically,

$$0.05 \times 0.99 = 0.0495, \qquad 0.95 \times 0.10 = 0.095,$$

so

$$P(B) = 0.0495 + 0.095 = 0.1445.$$

This calculation is an example of the law of total probability: we express the probability of B as a sum over different scenarios (A and A^c) weighted by their prior probabilities and the corresponding conditional probabilities of B.

Finally, suppose the radar has reported a detection. We would like to know how likely it is that there really is an airplane in the sky, that is, we want $P(A \mid B)$. By the definition of conditional probability,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

We already computed both the numerator and the denominator:

$$P(A \cap B) = 0.0495, \qquad P(B) = 0.1445,$$

so

$$P(A \mid B) = \frac{0.0495}{0.1445} \approx 0.34.$$

Thus, even though the radar is quite reliable in both scenarios, the probability that there is actually an airplane given that the radar reports a detection is only about 34%.
This example illustrates three basic tools that arise from conditional probabilities: the product rule, which expresses joint probabilities in terms of conditional probabilities; the law of total probability, which decomposes a probability into contributions from different scenarios; and the use of conditional probability to update our beliefs about A when we observe B. Variants of these ideas will appear repeatedly in more complex models.

1.2.2 The multiplication rule

Conditional probabilities lead to a general multiplication rule that expresses the probability of several events occurring together as a product of ordinary and conditional probabilities.
We start from the definition of conditional probability. For events A and B with $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Rearranging this equality gives

$$P(A \cap B) = P(B)\, P(A \mid B).$$

This says that the probability that both A and B occur is equal to the probability that B occurs, times the conditional probability that A occurs given that B has occurred. Since we are free to choose which event we treat as "first", we can also write

$$P(A \cap B) = P(A)\, P(B \mid A).$$

These are the simplest forms of the multiplication rule and they are exactly what we used in the radar example when we computed the probability of a particular path in the tree diagram.
The same idea extends to three events. Consider events A, B, and C. We can write their intersection as

$$A \cap B \cap C = (A \cap B) \cap C.$$

Applying the multiplication rule to the "composite" event $A \cap B$ and the event C gives

$$P(A \cap B \cap C) = P(A \cap B)\, P\big(C \mid A \cap B\big).$$

We can then expand $P(A \cap B)$ using the two-event rule:

$$P(A \cap B) = P(A)\, P(B \mid A).$$

Substituting into the previous expression yields

$$P(A \cap B \cap C) = P(A)\, P(B \mid A)\, P\big(C \mid A \cap B\big).$$

This formula has a natural interpretation in terms of a three-stage tree: first we pick A or A^c, then from there we pick B or B^c, and finally we pick C or C^c. The probability of a leaf is the product of the probability of the first branch, the conditional probability of the second branch given the first, and the conditional probability of the third branch given the previous two.
The same reasoning works for more than three events. Let $A_1, A_2, \ldots, A_n$ be events and assume that all the conditional probabilities we write below are well-defined. We can write

$$A_1 \cap A_2 \cap \cdots \cap A_n = (A_1 \cap \cdots \cap A_{n-1}) \cap A_n,$$

and apply the multiplication rule step by step. The result is the general multiplication rule:

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \prod_{i=2}^{n} P\big(A_i \mid A_1 \cap \cdots \cap A_{i-1}\big).$$

In words, the probability that all of $A_1, \ldots, A_n$ occur is equal to the probability that A_1 occurs, times the probability that A_2 occurs given that A_1 has occurred, times the probability that A_3 occurs given that A_1 and A_2 have occurred, and so on. This is the most general form of the multiplication rule and it underlies the computation of path probabilities in multi-stage models.

1.2.3 Total probability theorem

The total probability theorem formalizes the "divide and conquer" idea: to find the probability of an event that can occur under several disjoint scenarios, we add its probabilities across those scenarios.
Consider a sample space Ω and events A_1, A_2, A_3 that form a **partition** of Ω. This means: - the events are pairwise disjoint, - their union is the whole sample space, - and each has positive probability:

$$A_i \cap A_j = \emptyset \text{ for } i \neq j, \qquad A_1 \cup A_2 \cup A_3 = \Omega, \qquad P(A_i) > 0.$$

We can think of A_1, A_2, A_3 as three mutually exclusive "scenarios" under which the experiment may unfold.
Now take an event B. The event B can only occur through one of these scenarios. In set notation,

$$B = (B \cap A_1) \cup (B \cap A_2) \cup (B \cap A_3).$$

The three sets on the right-hand side are disjoint, because the A_i are disjoint. Therefore, by the additivity axiom,

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3).$$

For each term we can use the multiplication rule:

$$P(B \cap A_i) = P(A_i)\, P(B \mid A_i), \qquad i = 1, 2, 3.$$

Substituting these expressions into the previous equation gives

$$P(B) = P(A_1)\, P(B \mid A_1) + P(A_2)\, P(B \mid A_2) + P(A_3)\, P(B \mid A_3).$$

This is the **total probability theorem** for three scenarios. It says that the total probability of B is the sum over all scenarios of:

(probability of the scenario) $\times$ (probability of B within that scenario).

Equivalently, $P(B)$ is a weighted average of the conditional probabilities $P(B \mid A_i)$, with weights $P(A_i)$ that sum to 1.
The same reasoning applies to any finite number of scenarios. If $A_1, \ldots, A_n$ form a partition of Ω and $P(A_i) > 0$ for all i, then for any event B,

$$P(B) = \sum_{i=1}^{n} P(A_i \cap B) = \sum_{i=1}^{n} P(A_i)\, P(B \mid A_i).$$

This is the general statement of the total probability theorem. If we partition Ω into countably many events $A_1, A_2, \ldots$, the same derivation goes through, except that we use countable additivity instead of finite additivity, and we obtain an infinite series

$$P(B) = \sum_{i=1}^{\infty} P(A_i)\, P(B \mid A_i).$$

In all cases, the key idea is to decompose B into disjoint pieces corresponding to different scenarios and then sum their probabilities.

1.2.4 Bayes' rule

Bayes' rule describes how to update the probabilities of different scenarios after observing an event.
We keep the same setting as for the total probability theorem. The sample space Ω is partitioned into disjoint events $A_1, A_2, \ldots, A_n$, which we think of as *scenarios*. They satisfy

$$A_i \cap A_j = \emptyset \text{ for } i \neq j, \qquad \bigcup_{i=1}^{n} A_i = \Omega, \qquad P(A_i) > 0.$$

The numbers $P(A_i)$ represent our *initial beliefs* about how likely each scenario is before seeing any additional information. There is also an event B that may occur under each scenario, and for each i we know the conditional probability $P(B \mid A_i)$, that is, how likely B is to occur if scenario A_i is the true one.
Once we perform the experiment, we may observe that B has actually occurred. This new information should cause us to revise our beliefs about which scenario A_i is more likely. The updated belief about scenario A_i is the conditional probability $P(A_i \mid B)$.
To compute $P(A_i \mid B)$, we start from the definition of conditional probability:

$$P(A_i \mid B) = \frac{P(A_i \cap B)}{P(B)}, \qquad P(B) > 0.$$

The numerator $P(A_i \cap B)$ can be written using the multiplication rule:

$$P(A_i \cap B) = P(A_i)\, P(B \mid A_i).$$

The denominator $P(B)$ is given by the total probability theorem:

$$P(B) = \sum_{j=1}^{n} P(A_j \cap B) = \sum_{j=1}^{n} P(A_j)\, P(B \mid A_j).$$

Substituting these expressions into the definition yields

$$P(A_i \mid B) = \frac{P(A_i)\, P(B \mid A_i)}{\sum_{j=1}^{n} P(A_j)\, P(B \mid A_j)}.$$

This formula is **Bayes' rule**. It combines three ingredients: - the prior probabilities $P(A_i)$, which encode our initial beliefs about each scenario; - the likelihoods $P(B \mid A_i)$, which tell us how compatible the observation B is with each scenario; - the normalizing denominator $\sum_j P(A_j)\, P(B \mid A_j)$, which ensures that the updated probabilities $P(A_i \mid B)$ sum to 1.
Conceptually, Bayes' rule starts from conditional probabilities "in one direction", $P(B \mid A_i)$, provided by the model, and uses them to compute conditional probabilities in the **reverse direction**, $P(A_i \mid B)$, which represent our revised beliefs about the scenarios after observing B. This is the basic mechanism of statistical inference: we begin with a probabilistic model and prior weights on possible explanations, observe data B, and then use Bayes' rule to update the probabilities of the different scenarios in light of the new evidence.

1.2.5 Independence

Independence formalizes the idea that learning one event has occurred does not change our beliefs about another event.
We return to the example of three tosses of a biased coin. The coin lands heads with probability p, where p is not necessarily $1/2$. The experiment can be represented by a tree with three stages, one for each toss. Each branch corresponds to a possible result at a given toss, and each leaf corresponds to a complete outcome of the three tosses. Probabilities and conditional probabilities are attached to the branches: for example, the probability of heads on the first toss is p, and the probability of heads on the second toss given that the first toss was heads is also p. Similarly, the probability of heads on the second toss given that the first toss was tails is p as well. More generally, in this model every toss has probability p of being heads, regardless of the results of previous tosses.
We can use the multiplication rule to compute the probability of any specific outcome. For instance, consider the outcome "tails, then heads, then tails". Its probability is obtained by multiplying along the corresponding path in the tree:

$$P(T, H, T) = (1-p)\, p\, (1-p).$$

We can also compute the probability of more complicated events. Take the event that there is *exactly one head* in the three tosses. This event consists of three distinct outcomes:

$$(H, T, T), \quad (T, H, T), \quad (T, T, H).$$

Using the tree, we find that each of these outcomes has probability $p(1-p)^2$. Therefore, the total probability of exactly one head is

$$P(\text{exactly one head}) = 3\, p(1-p)^2.$$

All three outcomes in this event are equally likely.
Now consider a conditional probability. Suppose we are told that there was exactly one head in the three tosses. We want the probability that the *first* toss was heads under this information. In other words, we want

$$P(H_1 \mid \text{exactly one head}).$$

Intuitively, within the event "exactly one head" the three possible outcomes are equally likely, so each should have conditional probability $1/3$. Only one of them—(H, T, T)—has heads on the first toss, so we expect

$$P(H_1 \mid \text{exactly one head}) = \frac{1}{3}.$$

We can verify this formally using the definition of conditional probability. Let E be the event "exactly one head" and G be the event "first toss is heads". Then

$$P(G \mid E) = \frac{P(G \cap E)}{P(E)}.$$

The event $G \cap E$ corresponds to the single outcome (H, T, T), which has probability $p(1-p)^2$. The denominator $P(E)$ is the probability of exactly one head, which we already computed as $3p(1-p)^2$. Thus

$$P(G \mid E) = \frac{p(1-p)^2}{3p(1-p)^2} = \frac{1}{3}.$$

This example has a special property: knowing the result of one toss does not change the probabilities for future tosses. For instance, the conditional probability of heads on the second toss given heads on the first toss is

$$P(H_2 \mid H_1) = p,$$

and the conditional probability of heads on the second toss given tails on the first toss is

$$P(H_2 \mid T_1) = p.$$

So the probability of heads on the second toss remains p, regardless of what happened on the first toss. We can also compute the unconditional probability of heads on the second toss using the total probability theorem:

$$P(H_2) = P(H_1)\, P(H_2 \mid H_1) + P(T_1)\, P(H_2 \mid T_1) = p \cdot p + (1-p) \cdot p = p.$$

Therefore, the unconditional probability of heads on the second toss is equal to both conditional probabilities. Learning the outcome of the first toss does not change our belief about the second toss.
This motivates the notion of independence. Let A and B be two events. Informally, B is independent of A if knowing that A has occurred does not change our belief about B, in the sense that

$$P(B \mid A) = P(B).$$

Using the multiplication rule,

$$P(A \cap B) = P(A)\, P(B \mid A).$$

If we also have $P(B \mid A) = P(B)$, then this becomes

$$P(A \cap B) = P(A)\, P(B).$$

This identity turns out to be the most convenient formal definition of independence. We say that two events A and B are *independent* if

$$P(A \cap B) = P(A)\, P(B).$$

This definition has several advantages. It is symmetric in A and B, so it does not distinguish between "A is independent of B" and "B is independent of A". It is also consistent with the intuitive definition: if $P(A) > 0$ and $P(B) > 0$, then

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = P(B),$$

and similarly $P(A \mid B) = P(A)$ whenever $P(B) > 0$. Finally, the product definition makes sense even when one of the events has probability zero, whereas conditional probabilities are not defined in that case.
It is important to distinguish independence from disjointness. Suppose A and B are disjoint events with strictly positive probabilities. Then

$$P(A \cap B) = 0,$$

but

$$P(A)\, P(B) > 0.$$

So the defining equality for independence does *not* hold, and A and B are not independent. Intuitively, disjoint events are as dependent as possible: if you know that A has occurred, you are certain that B has not, and vice versa. In contrast, independent events are such that the occurrence of one does not affect the probability of the other.
In many real-world situations, independence is a reasonable modeling assumption when two events are determined by physically separate and noninteracting processes. For example, the result of a coin toss and the weather on a specific future day can often be modeled as independent. In other cases, apparent independence may arise from a numerical coincidence, so one should always think carefully about whether the independence assumption is justified in a given context.

1.2.6 Independence of event complements

Independence between events automatically extends to their complements in a natural way.
Suppose that events A and B are independent. Intuitively, if learning that A occurred does not change our belief about whether B will occur, it also should not change our belief about whether B will *not* occur. In other words, if A is independent of B, then it is reasonable to expect that A is also independent of B^c. We now show this formally.
Start with the event A. We can decompose A into two disjoint pieces:

$$A = (A \cap B) \cup (A \cap B^c).$$

These two sets are disjoint, and together they make up the whole of A. By additivity,

$$P(A) = P(A \cap B) + P(A \cap B^c).$$

Since A and B are independent, their joint probability factorizes:

$$P(A \cap B) = P(A)\,P(B).$$

Substituting this into the previous equation gives

$$P(A) = P(A)\,P(B) + P(A \cap B^c).$$

Solving for $P(A \cap B^c)$ yields

$$P(A \cap B^c) = P(A) - P(A)\,P(B) = P(A)\big(1 - P(B)\big).$$

Using the relation $P(B^c) = 1 - P(B)$, we can rewrite this as

$$P(A \cap B^c) = P(A)\,P(B^c).$$

This is exactly the condition for A and B^c to be independent. Thus, we have shown that if A and B are independent, then A and B^c are independent as well.

1.2.7 Conditional independence

Conditional independence extends the idea of independence to a setting where we already condition on some additional information.
Conditional probabilities are like ordinary probabilities, except that they apply in a new situation where we are told that a certain event has occurred. Because of this, any notion we define for ordinary probabilities has an analogue for conditional probabilities. In particular, we can talk about conditional independence, which is simply the notion of independence applied to a conditional model.
More concretely, suppose we have events A, B, and C in a probability space, with $P(C) > 0$. Once we are told that C has occurred, we work with the conditional probabilities $P(\cdot \mid C)$. In the original (unconditional) model, independence of A and B is defined by

$$P(A \cap B) = P(A)\,P(B).$$

In the conditional model given C, we use exactly the same pattern but with conditional probabilities in place of ordinary ones. We say that A and B are *conditionally independent given* C if

$$P(A \cap B \mid C) = P(A \mid C)\,P(B \mid C),$$

provided all these conditional probabilities are defined. Intuitively, this means that after we learn that C has occurred, learning whether A occurs does not change our belief about B, and learning whether B occurs does not change our belief about A, within the "universe" where C holds.
A natural question is whether ordinary independence implies conditional independence. The answer is no in general. It may happen that A and B are independent in the unconditional model, so that $P(A \cap B) = P(A)\,P(B)$, but once we restrict attention to a particular event C, the geometry of the sets inside C changes. For example, it could be that within C the events A and B have no overlap at all, so that

$$P(A \cap B \mid C) = 0,$$

while both $P(A \mid C)$ and $P(B \mid C)$ are positive. In that case,

$$P(A \cap B \mid C) \neq P(A \mid C)\,P(B \mid C),$$

and A and B are not conditionally independent given C. In this way, conditioning on C can introduce dependencies even when none were present before.
The main takeaway is that conditional independence is defined by the same product relation as ordinary independence, but using conditional probabilities. However, unconditional independence does not guarantee conditional independence once we restrict to a particular event C.

1.2.8 Independence vs conditional independence

Independence and conditional independence can behave very differently; independent events may become dependent after conditioning.
Consider two possible coins, coin A and coin B. If we know that coin A has been chosen, then in this conditional universe every toss is heads with probability 0.9, and different tosses are independent. Formally, given "coin A", the coin flips are *conditionally independent* and identically distributed with $P(H) = 0.9$. Similarly, given "coin B", the flips are conditionally independent with $P(H) = 0.1$ at each toss.
Now suppose that, before tossing, we pick one of the two coins at random, with

$$P(\text{coin A}) = 0.5, \qquad P(\text{coin B}) = 0.5,$$

and then we keep tossing that same coin. We consider the *overall* model where we do not know in advance which coin was selected, and we ask whether the different tosses are independent in this unconditional model.
First compute the unconditional probability of heads on the 11th toss. Using the total probability theorem,

$$P(H_{11}) = P(\text{coin A})\, P(H_{11} \mid \text{coin A}) + P(\text{coin B})\, P(H_{11} \mid \text{coin B}) = 0.5 \cdot 0.9 + 0.5 \cdot 0.1 = 0.5.$$

So before seeing any outcomes, each toss has probability 0.5 of being heads on average.
Next, suppose we are told that the first 10 tosses all resulted in heads. Intuitively, this makes coin A extremely likely: observing 10 heads in a row is very plausible under coin A, but extremely unlikely under coin B. In fact, by Bayes' rule one can show that

$$P(\text{coin A} \mid H_1, \ldots, H_{10})$$

is very close to 1, and the conditional probability of heads on the 11th toss satisfies

$$P(H_{11} \mid H_1, \ldots, H_{10}) \approx P(H_{11} \mid \text{coin A}) = 0.9.$$

Thus

$$P(H_{11} \mid H_1, \ldots, H_{10}) \approx 0.9 \neq 0.5 = P(H_{11}).$$

Information about the first ten tosses changes our belief about the 11th toss, so the 11th toss is *not* independent of the first ten in the overall model.
This example shows that even if the tosses are conditionally independent given the coin (they are independent within each conditional model), they need not be independent in the unconditional model where the underlying coin is unknown. Conditional independence does not imply unconditional independence, and unconditional independence does not imply conditional independence.

1.2.9 Independence of a collection of events

When we talk about independence for many events, we want a notion that captures "no information flow" between any subset and the rest of the family.
Suppose we have a collection of events $A_1, A_2, A_3, \ldots$. Intuitively, we would like to say that this family is independent if knowing what happened for some of the events does not change our beliefs about the others. For example, if A_1, A_2, A_3 are independent, then learning that A_1 occurred and A_2 did not occur should not change the probability we assign to A_3; the conditional probability of A_3 under that information should still be $P(A_3)$. More generally, any information about which of A_1, A_2 occurred should leave our probability for A_3 unchanged.
Stating independence directly in terms of "any information about some events does not affect the others" would lead to a very complicated definition. Instead, we use a cleaner condition that parallels the two-event case. Recall that two events A and B are independent exactly when

$$P(A \cap B) = P(A)\, P(B).$$

We extend this pattern to collections by requiring that we can always compute probabilities of intersections by multiplying individual probabilities.
For concreteness, consider three events A_1, A_2, A_3. We say that they are *independent* if the following relations all hold:

$$P(A_1 \cap A_2) = P(A_1)\, P(A_2),$$

$$P(A_1 \cap A_3) = P(A_1)\, P(A_3),$$

$$P(A_2 \cap A_3) = P(A_2)\, P(A_3),$$

and, in addition,

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)\, P(A_2)\, P(A_3).$$

The first three conditions mean that every pair of events is independent; this is called *pairwise independence*. The last condition is extra and does not follow automatically from pairwise independence. It enforces that the three-way intersection probability also factors as a product of individual probabilities. For three events, our definition requires both pairwise independence and the three-way product condition.
This definition extends to any finite family $A_1, \ldots, A_n$. We say that these events are independent if for every choice of distinct indices $i_1, \ldots, i_k$ (with $k \geq 2$) we have

$$P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k}).$$

In other words, the intersection of any subcollection of events has probability equal to the product of the individual probabilities. One can prove that, when these conditions are satisfied, all the intuitive "no-information" properties follow. For example, if A_1, A_2, A_3 are independent in this sense, then for any pattern of occurrences of A_1 and A_2 we have

$$P(A_3 \mid \text{any specification of } A_1, A_2) = P(A_3),$$

so knowing which of A_1 and A_2 occurred does not change the probability of A_3. The formal proof that the product conditions imply all such conditional equalities is somewhat tedious, so it is usually omitted, but the key point is that the product definition compactly captures the idea that information about any subfamily of events does not affect the probabilities of the remaining events.

1.2.10 Independence vs pairwise independence

Independence of a collection of events is stronger than pairwise independence; events can be pairwise independent without being independent as a family.
Consider two independent tosses of a fair coin. Let H_1 be the event that the first toss is heads, and H_2 the event that the second toss is heads. Each toss has probability 1/2 of heads, and the independence assumption gives

$$P(H_1 \cap H_2) = P(H_1)\, P(H_2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

By symmetry, each of the four outcomes (H, H), (H, T), (T, H), and (T, T) has probability 1/4.
Now define a new event C as "the two tosses have the same result", which corresponds to the outcomes (H, H) and (T, T). Thus

$$P(C) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

We first check pairwise independence. Take H_1 and C:

$$H_1 \cap C$$

is the event that the first toss is heads and the two tosses are the same, which happens only for (H, H), so

$$P(H_1 \cap C) = \frac{1}{4}.$$

On the other hand,

$$P(H_1)\, P(C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

So $P(H_1 \cap C) = P(H_1)\, P(C)$, and H_1 and C are independent. By symmetry, H_2 and C are also independent, and we already know that H_1 and H_2 are independent. Thus all three pairs (H_1, H_2), (H_1, C), and (H_2, C) are pairwise independent.
To test whether H_1, H_2, and C are independent as a collection, we must also look at the three-way intersection. The event

$$H_1 \cap H_2 \cap C$$

means "first toss heads, second toss heads, and the two tosses are the same", which again corresponds only to the outcome (H, H), so

$$P(H_1 \cap H_2 \cap C) = \frac{1}{4}.$$

The product of individual probabilities is

$$P(H_1)\, P(H_2)\, P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Since $\frac{1}{4} \neq \frac{1}{8}$, the factorization condition for three events fails. Therefore H_1, H_2, C are not independent as a collection, even though every pair among them is independent. This illustrates that pairwise independence does not imply independence of the whole family.

The intuition is that knowing only H_1 does not help us decide whether C occurs: given that the first toss is heads, the second toss is equally likely to be heads or tails, so

$$P(C \mid H_1) = \frac{1}{2} = P(C).$$

Similarly $P(C \mid H_2) = P(C)$. However, if we know both H_1 and H_2 occurred, then we know the results of the two tosses are the same, so

$$P(C \mid H_1 \cap H_2) = 1 \neq P(C).$$

Thus the pair (H_1, H_2) together carries information about C, even though each event separately does not. This is exactly the situation where pairwise independence holds but full independence of the collection fails.

1.2.11 Reliability

Independence lets us analyze complex systems by breaking them into simpler independent pieces, which is especially useful in reliability calculations.
Consider a system with n components (or units). The i-th unit can be "up" (working) or "down" (failed). Let U_i be the event that unit i is up, with

$$P(U_i) = p_i,$$

and assume the events $U_1, \ldots, U_n$ are independent. Intuitively, this means that whether some units are up or down does not change the probabilities for the others. Equivalently, if we define $F_i = U_i^c$ as the event that unit i has failed, then the failure events $F_1, \ldots, F_n$ are also independent: information about which units have failed does not affect the failure probabilities of the remaining units.
Let us look at a system with three components and define the event "system up" to mean that there exists a path from left to right consisting only of working units.
First, consider a series system where all three units lie on the same path. In this case, the system is up if and only if all three units are up. The event "system up" is

$$U_1 \cap U_2 \cap U_3,$$

and by independence its probability is

$$P(\text{system up}) = P(U_1 \cap U_2 \cap U_3) = P(U_1)\, P(U_2)\, P(U_3) = p_1 p_2 p_3.$$

Next, consider a parallel system where three units provide alternative paths from left to right. The system is up as long as at least one unit is up. The event "system up" is

$$U_1 \cup U_2 \cup U_3.$$

Independence is most convenient for intersections, so we rewrite this probability using complements. The complement of "at least one unit is up" is "all units are down", i.e.

$$(U_1 \cup U_2 \cup U_3)^c = F_1 \cap F_2 \cap F_3.$$

Therefore,

$$P(\text{system up}) = P(U_1 \cup U_2 \cup U_3) = 1 - P(F_1 \cap F_2 \cap F_3).$$

By independence of the failure events, this intersection probability factors:

$$P(F_1 \cap F_2 \cap F_3) = P(F_1)\, P(F_2)\, P(F_3).$$

Since $P(F_i) = 1 - p_i$, we obtain

$$P(\text{system up}) = 1 - (1 - p_1)(1 - p_2)(1 - p_3).$$

These formulas illustrate how independence allows us to compute the reliability (probability of correct operation) of systems built from multiple components. For series configurations we multiply the success probabilities of all components, and for parallel configurations we use complements and multiply failure probabilities. More complex networks can be analyzed by combining these ideas.

EXAMPLE: THE KING'S SIBLING

The probability that the king's sibling is female depends on how we model the royal family's childbearing decision. Assume first that the royal couple decided in advance to have exactly two children, that each child is independently a boy or a girl with probability 1/2, and that a boy always becomes king if there is at least one boy. Under these assumptions there are four equally likely gender outcomes for the two children:

$$(B, B),\ (B, G),\ (G, B),\ (G, G),$$

each with probability 1/4. Since there is a king, there must be at least one boy, so the outcome (G, G) is impossible. Conditioning on "at least one boy", we restrict to

$$(B, B),\ (B, G),\ (G, B),$$

which are now equally likely and each has conditional probability 1/3. In two of these three outcomes the sibling is a girl, so

$$P(\text{sibling is female} \mid \text{there is a king, with exactly two children}) = \frac{2}{3}.$$

However, this answer relies crucially on the assumption that "two children" was fixed in advance. If we change the family rule, the answer changes. For example, if the royal family "has children until the first boy is born," then the fact that there are two children tells us the first was a girl and the second a boy, so the sibling is certainly a girl and the probability is 1. If instead they "have children until they get two boys," then being told there are exactly two children implies both are boys, so the sibling cannot be a girl and the probability is 0.

Thus 2/3 is the correct answer only under the model where the number of children (two) is fixed beforehand, genders are independent and equally likely, and a boy always becomes king. Different modeling assumptions lead to different probabilities, which illustrates how important it is to clarify the probabilistic model before computing.

1.3 Counting

1.3.1 The counting principle

We often construct an outcome through a sequence of stages, making a choice at each stage. The *counting* principle says: if there are r stages and at stage i we always have n_i options (independent of previous choices), then the total number of possible outcomes is

$$n_1 n_2 \cdots n_r.$$

EXAMPLE: OUTFITS

Suppose you have 4 shirts, 3 ties, and 2 jackets. Think of getting dressed as a 3-stage process:

shirt : 4 choices, tie : 3 choices, jacket : 2 choices.

By the counting principle, the number of different outfits is

$$4 \cdot 3 \cdot 2 = 24.$$

Figure 1: Enter Caption

EXAMPLE: LICENSE PLATES
Consider license plates of the form "2 letters followed by 3 digits." We construct the plate sequentially.
With repetition allowed. At each stage:

$$\underbrace{26}_{\text{1st letter}} \cdot \underbrace{26}_{\text{2nd letter}} \cdot \underbrace{10}_{\text{1st digit}} \cdot \underbrace{10}_{\text{2nd digit}} \cdot \underbrace{10}_{\text{3rd digit}}.$$

So the number of such plates is

$$26^2 \cdot 10^3.$$

Without repetition. Now require that no letter and no digit can be used more than once. Then:

$$\underbrace{26}_{\text{1st letter}} \cdot \underbrace{25}_{\text{2nd letter}} \cdot \underbrace{10}_{\text{1st digit}} \cdot \underbrace{9}_{\text{2nd digit}} \cdot \underbrace{8}_{\text{3rd digit}}.$$

Thus, the number of such plates is

$$26 \cdot 25 \cdot 10 \cdot 9 \cdot 8.$$

EXAMPLE: PERMUTATIONS AND FACTORIAL
Let a set have n distinct elements; we want to order them (form a permutation). Think of n slots and fill them one at a time:

$$\underbrace{n}_{\text{1st slot}} \cdot \underbrace{(n-1)}_{\text{2nd slot}} \cdot \underbrace{(n-2)}_{\text{3rd slot}} \cdots \underbrace{1}_{\text{last slot}}.$$

The total number of permutations is

$$n! = n(n-1)(n-2)\cdots 2 \cdot 1.$$

EXAMPLE: NUMBER OF SUBSETS
Let a set have n elements; we want to count its subsets. Consider the elements one by one. For each element, we have 2 options: include it in the subset or not. By the counting principle, the total number of subsets is

$$2 \cdot 2 \cdot \cdots \cdot 2 = 2^n,$$

with n factors.
Sanity check for $n = 1$. If the set is $\{a\}$, its subsets are

$$\{a\}, \quad \varnothing,$$

so there are 2 subsets, in agreement with $2^1 = 2$. Note that when we count subsets we include both the whole set and the empty set.

1.3.2 A die-roll example

We roll an ordinary six-sided die six times, and we are interested in the probability that all six results are different; denote this event by A. We assume that all 6^6 possible sequences of six rolls are equally likely, so we are in a discrete uniform model and probabilities can be computed by counting.

Sample space
A typical outcome is a sequence such as $(2, 3, 4, 3, 6, 2)$. At each of the 6 rolls we have 6 possible results, so by the counting principle the total number of outcomes is

$$|\Omega| = 6^6.$$

Under the equally likely assumption, each sequence has probability $1/6^6$. This is consistent with viewing the experiment as 6 independent rolls of a fair die, each outcome having probability $(1/6)^6 = 1/6^6$.

Counting the favorable outcomes
Event A consists of all sequences in which no number is repeated, i.e., each of the numbers $1, \ldots, 6$ appears exactly once. Such sequences are precisely the permutations of $\{1, 2, 3, 4, 5, 6\}$, and there are

$$|A| = 6!$$

of them.

Probability of all distinct rolls
By the discrete uniform model,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{6!}{6^6}.$$

1.3.3 Combinations

We start with a set of n elements and a non-negative integer k. A *combination* is a subset of the original set that has exactly k elements. The number of such subsets is denoted by the binomial coefficient

$$\binom{n}{k},$$

read as "n-choose-k".

Two ways to count ordered k-tuples
Consider a related problem: construct an ordered list of k *distinct* elements chosen from the n-element set.
First method (direct counting). Think of k slots and fill them one by one with distinct elements:

$$n \cdot (n-1) \cdot (n-2) \cdots (n-k+1).$$

This product can be written as

$$n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

Second method (choose then order). First choose which k elements will appear (ignore order), then order them.

$$\underbrace{\binom{n}{k}}_{\text{choose the set of } k \text{ elements}} \cdot \underbrace{k!}_{\text{permute these } k \text{ elements}}.$$

Both methods count the same set of ordered lists, so

$$\binom{n}{k} k! = \frac{n!}{(n-k)!}.$$

Solving for $\binom{n}{k}$ gives the standard formula

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!},$$

valid for integers $n \geq 0$ and $0 \leq k \leq n$.

Extreme cases and the convention $0! = 1$
Consider $\binom{n}{n}$. From the combinatorial definition, choosing n elements from an n-element set leaves only one possibility: take all elements, so $\binom{n}{n} = 1$. From the formula,

$$\binom{n}{n} = \frac{n!}{n!\,0!},$$

so this equals 1 if we adopt the convention $0! = 1$.
Now consider $\binom{n}{0}$. Combinatorially, there is exactly one subset with zero elements: the empty set, so $\binom{n}{0} = 1$. From the formula,

$$\binom{n}{0} = \frac{n!}{0!\,n!} = 1,$$

again consistent with $0! = 1$.

A useful binomial identity
The sum

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}$$

can be given a combinatorial interpretation. Here, $\binom{n}{k}$ counts the subsets of size k of an n-element set. Summing over all k from 0 to n counts all subsets of the set. But the total number of subsets of an n-element set is 2^n. Therefore,

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

1.3.4 Binomial probabilities

The binomial coefficients $\binom{n}{k}$ are closely related to probabilities in repeated coin-tossing experiments. Consider a coin tossed n times independently. Let p be the probability of heads on a single toss, so the probability of tails is $1 - p$. We want the probability of observing exactly k heads in these n tosses.

Probability of a particular sequence
Take a concrete example with $n = 6$, and consider a specific sequence of heads and tails, e.g. H T T H H H. By independence, the probability of this sequence is

$$p \cdot (1-p) \cdot (1-p) \cdot p \cdot p \cdot p = p^4(1-p)^2.$$

In general, for any *given* sequence with k heads and $n - k$ tails, the probability of that sequence is

$$p^k(1-p)^{n-k},$$

because each head contributes a factor p and each tail a factor $1 - p$.

Number of k-head sequences
We now want the probability of the event "exactly k heads in n tosses". This event consists of all sequences of length n that contain exactly k heads. Every such sequence has the same probability $p^k(1-p)^{n-k}$. Therefore, the desired probability equals

$$\big(\text{number of sequences with exactly } k \text{ heads}\big) \times p^k(1-p)^{n-k}.$$

To count the sequences with exactly k heads, think of the n tosses as n time slots. To specify a k-head sequence, we must choose which k of the n slots contain heads; the remaining $n - k$ slots will contain tails. Thus, the number of such sequences is

$$\binom{n}{k}.$$

The binomial formula
Putting everything together, the probability of exactly k heads in n independent tosses of a coin with head probability p is

$$P(\text{exactly } k \text{ heads}) = \binom{n}{k} p^k(1-p)^{n-k},$$

which is called the *binomial* probability formula.

1.3.5 Example: Each person gets an ace

We have a standard 52-card deck, dealt fairly to four players, 13 cards each (as in bridge). Outcomes are all partitions of the 52 cards into four hands of size 13, and all such partitions are assumed equally likely. We want the probability that each player gets *exactly one ace*.

Counting via multinomial coefficients
The sample space size is the number of ways to partition 52 distinct cards into hands of sizes $13, 13, 13, 13$, namely the multinomial coefficient

$$|\Omega| = \binom{52}{13, 13, 13, 13} = \frac{52!}{13!\,13!\,13!\,13!}.$$

Now count the favorable outcomes where each person gets exactly one ace.
First distribute the four aces: there are

$$4 \cdot 3 \cdot 2 \cdot 1 = 4!$$

ways to assign the four distinct aces to the four distinct players (each player gets one).
Then distribute the remaining 48 non-aces arbitrarily, giving 12 additional cards to each player. The number of such partitions is

$$\binom{48}{12, 12, 12, 12} = \frac{48!}{12!\,12!\,12!\,12!}.$$

By the counting principle, the number of favorable outcomes is

$$|A| = 4! \cdot \binom{48}{12, 12, 12, 12} = 4! \cdot \frac{48!}{12!\,12!\,12!\,12!}.$$

Thus, the desired probability is

$$P(A) = \frac{|A|}{|\Omega|} = \frac{4! \cdot \dfrac{48!}{12!\,12!\,12!\,12!}}{\dfrac{52!}{13!\,13!\,13!\,13!}}.$$

Alternative "aces first" viewpoint
There is a faster way to obtain the same probability. Imagine the 52 slots corresponding to the four players' 13-card hands. Place the four aces *first*, one at a time, uniformly at random among the 52 slots.
The first ace can go anywhere.
For the second ace, there are 51 remaining slots, of which 39 belong to players who do not yet have an ace (3 players $\times$ 13 slots each). So the conditional probability that the second ace goes to a different player than the first is

$$\frac{39}{51}.$$

Assuming this has happened, two players now have aces. For the third ace there are 50 remaining slots; the "good" slots are those belonging to the two players without aces, i.e. $2 \times 13 = 26$ slots. So the corresponding probability is

$$\frac{26}{50}.$$

Assuming all three aces have gone to different players, three players now have aces. For the fourth ace, there are 49 remaining slots, of which 13 belong to the last player without an ace, giving probability

$$\frac{13}{49}.$$

Multiplying these conditional probabilities, the overall probability that the four aces land in slots belonging to four different players is

$$P(A) = \frac{39}{51} \cdot \frac{26}{50} \cdot \frac{13}{49}.$$

One can check algebraically that this expression is equal to the earlier ratio of multinomial coefficients.

1.4 Discrete random variables

1.4.1 Definition of random variables

A random variable is a function from the sample space to the real numbers. It assigns to each outcome of the experiment a numerical value, determined once the outcome is known.

Example: Weight of a student
Let the sample space be $\Omega = \{a, b, c, d\}$, where each element is a student. The experiment is to pick one student at random and record their weight in kilograms. Define $W(\omega)$ as the weight of student ω. Then W is a random variable, because for each outcome $\omega \in \Omega$ it produces a real number $W(\omega)$.

Another random variable and a function of them
On the same experiment, define $H(\omega)$ as the height (in meters) of student ω. This H is also a random variable. We can form a new random variable by combining them, for instance the body mass index

$$B(\omega) = \frac{W(\omega)}{H(\omega)^2}.$$

Once the outcome ω is known, both $W(\omega)$ and $H(\omega)$ are known, so $B(\omega)$ is determined as well.

Discrete and continuous random variables
Random variables can be discrete or continuous. For example, the number of heads in 10 coin tosses is a discrete random variable taking values in $\{0, 1, \ldots, 10\}$. The time at which an event occurs, measured with infinite accuracy, is modeled as a continuous random variable taking real values in some interval.

Notation
We use uppercase letters such as X, Y, W, H for random variables (the functions) and lowercase letters such as x, y, w, h for their numerical values. If X and Y are random variables, then $X + Y$ is also a random variable, defined by

$$(X + Y)(\omega) = X(\omega) + Y(\omega).$$

In general, any function of random variables is again a random variable.

1.4.2 Probability mass function

A discrete random variable can take different numerical values, and some of these values are more likely than others. These relative likelihoods are described by the *probability mass function* (PMF), which gives the probability of each possible numerical value.

Definition of the PMF
Let X be a discrete random variable. For any real number x, we write

$$p_X(x) = \mathbb{P}(X = x).$$

This is the probability of the event that X takes the value x. The function p_X is called the PMF (or probability law, or probability distribution) of X.
The argument x ranges over the possible values that X can take. For each such x, $p_X(x)$ is a number between 0 and 1. If we had another random variable Y on the same sample space, its PMF would be denoted $p_Y(y) = \mathbb{P}(Y = y)$.

Basic properties
The PMF satisfies two fundamental properties:

$$p_X(x) \geq 0 \quad \text{for all } x,$$
$$\sum_x p_X(x) = 1,$$

where the sum is over all possible values of X. The first property holds because probabilities are never negative. The second holds because the events $\{X = x\}$ for all possible x are disjoint and together cover the entire sample space, so their probabilities must add up to 1.

Example: A simple random variable
Consider an experiment with four equally likely outcomes a, b, c, d. Suppose a random variable X is defined by

$$X(a) = 5, \quad X(b) = 5, \quad X(c) = 4, \quad X(d) = 3.$$

Each outcome has probability 1/4.
The event $\{X = 5\}$ is $\{a, b\}$, so

$$p_X(5) = \mathbb{P}(X = 5) = \mathbb{P}(\{a, b\}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Similarly,

$$p_X(4) = \mathbb{P}(X = 4) = \mathbb{P}(\{c\}) = \frac{1}{4},$$
$$p_X(3) = \mathbb{P}(X = 3) = \mathbb{P}(\{d\}) = \frac{1}{4}.$$

For any other x, we have $p_X(x) = 0$. These values satisfy $p_X(3) + p_X(4) + p_X(5) = \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$, as required.

Example: Sum of two tetrahedral dice
Consider rolling two independent tetrahedral dice, each with faces $\{1, 2, 3, 4\}$. Let X be the result of the first roll and Y the result of the second roll. All 16 pairs (i, j) with $i, j \in \{1, 2, 3, 4\}$ are assumed equally likely, each with probability 1/16.
Define a new random variable

$$Z = X + Y.$$

The possible values of Z are $2, 3, 4, 5, 6, 7, 8$. To compute the PMF of Z, we proceed value by value, each time collecting the outcomes that give that sum and adding their probabilities.
For example, the event $\{Z = 2\}$ is $\{(1, 1)\}$, so

$$p_Z(2) = \mathbb{P}(Z = 2) = \frac{1}{16}.$$

The event $\{Z = 3\}$ is $\{(1, 2), (2, 1)\}$, so

$$p_Z(3) = \mathbb{P}(Z = 3) = \frac{2}{16}.$$

The event $\{Z = 4\}$ is $\{(1, 3), (2, 2), (3, 1)\}$, so

$$p_Z(4) = \mathbb{P}(Z = 4) = \frac{3}{16}.$$

Continuing in the same way, we find

$$p_Z(5) = \frac{4}{16}, \quad p_Z(6) = \frac{3}{16}, \quad p_Z(7) = \frac{2}{16}, \quad p_Z(8) = \frac{1}{16}.$$

These numbers are all nonnegative and they add up to 1:

$$\frac{1}{16} + \frac{2}{16} + \frac{3}{16} + \frac{4}{16} + \frac{3}{16} + \frac{2}{16} + \frac{1}{16} = 1.$$

Thus we have completely determined the probability mass function of Z.

1.4.3 Bernoulli and indicator random variables

A *Bernoulli* random variable is a discrete random variable that takes only the values 0 and 1. Its distribution is determined by a parameter $p \in [0, 1]$, interpreted as the probability of the outcome 1.
Using PMF notation, a Bernoulli random variable X satisfies

$$\mathbb{P}(X = 1) = p, \qquad \mathbb{P}(X = 0) = 1 - p.$$

If we plot its PMF, we obtain two bars: one at $x = 0$ of height $1 - p$, and one at $x = 1$ of height p. Bernoulli random variables model experiments with two possible outcomes, such as success/failure or heads/tails.

Indicator random variables
Let Ω be a sample space and let $A \subseteq \Omega$ be an event. The *indicator* random variable of A is defined by

$$I_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

Thus $I_A = 1$ exactly when the event A occurs, and $I_A = 0$ otherwise.
Since I_A takes only the values 0 and 1, it is a Bernoulli random variable. Moreover,

$$\mathbb{P}(I_A = 1) = \mathbb{P}(A), \qquad \mathbb{P}(I_A = 0) = 1 - \mathbb{P}(A),$$

so I_A is Bernoulli with parameter $p = \mathbb{P}(A)$. Indicator random variables are useful because they convert questions about events into questions about random variables, where algebraic manipulations are often easier.

1.4.4 Uniform random variables

A *discrete uniform* random variable is one whose possible values all have the same probability. It takes integer values in some finite interval and assigns equal mass to each of them.

Definition
Let a and b be integers with $a \leq b$. We consider the experiment of choosing an integer at random from the set

$$\{a, a+1, a+2, \ldots, b\}.$$

There are $b - a + 1$ possible values.
Let X be the random variable that equals the chosen integer. Then X has PMF

$$\mathbb{P}(X = x) = \begin{cases} \dfrac{1}{b-a+1} & \text{if } x \in \{a, a+1, \ldots, b\}, \\ 0 & \text{otherwise.} \end{cases}$$

In this setting, the outcome of the experiment is already a number, and this number is the value of the random variable, so there is no distinction between outcome and numerical value.

Example: Seconds on a digital clock
Consider the seconds reading on a digital clock. The possible values are the integers from 0 to 59, so there are 60 possibilities. If you glance at the clock at a time chosen completely at random, there is no reason to think one second value is more likely than another.
Let S be the seconds reading. Then S is (ideally) discrete uniform on $\{0, 1, \ldots, 59\}$, with

$$\mathbb{P}(S = s) = \frac{1}{60}, \quad s = 0, 1, \ldots, 59.$$

Degenerate case
If $a = b$, then the range of X consists of a single value a. In that case,

$$\mathbb{P}(X = a) = 1,$$

and X is a *constant* or *degenerate* random variable. It has no randomness in the usual sense, but it still fits the formal definition of a random variable as a function on the sample space.

1.4.5 Binomial random variables

A *binomial* random variable models the number of successes in a fixed number of independent, identical trials, where each trial has two outcomes (success/failure) and the same success probability p. Typical examples are repeated coin tosses or repeated yes/no experiments.

Definition
Consider n independent trials. On each trial, the probability of success is p and the probability of failure is $1 - p$. Define the random variable X to be the number of successes observed in these n trials.
Then X is called a *binomial* random variable with parameters n and p. Its possible values are $0, 1, \ldots, n$.

PMF of a binomial random variable
We now recall the probability that X takes a specific value k. The event $\{X = k\}$ consists of all sequences of n trials that contain exactly k successes and $n - k$ failures. Each particular sequence with k successes has probability

$$p^k (1-p)^{n-k},$$

because we multiply a factor p for each success and a factor $1 - p$ for each failure.
The number of such sequences is

$$\binom{n}{k},$$

since we must choose which k of the n positions are successes. Therefore, the PMF of X is

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

Interpretation and shape

When the coin is fair, $p = 1/2$, the PMF is largest near $k = n/2$. For example, with $n = 10$, the most likely outcome is $X = 5$, and probabilities decrease as k moves away from 5 in either direction. For larger n, such as $n = 100$, the most likely values of X form a band around np; values far from np have very small probability.

When p is small (for instance $p = 0.2$), small values of X are more likely. If n is moderate, such as $n = 10$, we typically see a few successes, often between 0 and 4. If n is large, such as $n = 100$ with $p = 0.1$, we say we expect "about" $np = 10$ successes, meaning that values near 10 (roughly in some neighborhood such as from 0 up to around 20) have noticeable probability, while much larger values are extremely unlikely.

1.4.6 Geometric random variables

A *geometric* random variable models the number of trials needed until the first success occurs, in a sequence of independent, identical trials with success probability p at each trial. Typical examples are repeated coin tosses until the first head appears, or repeated attempts at an experiment until it succeeds.

Definition and PMF

Consider an infinite sequence of independent trials, each with probability p of success and $1 - p$ of failure, where $0 < p \leq 1$. Let X be the number of the trial on which the first success occurs. Then X is called a *geometric* random variable. Its possible values are the positive integers $1, 2, 3, \ldots$.

To compute the PMF, fix $k \geq 1$. The event $\{X = k\}$ means that the first $k - 1$ trials are failures and the k-th trial is a success. By independence,

$$\mathbb{P}(X = k) = (1 - p)^{k-1} p, \qquad k = 1, 2, 3, \ldots$$

As k increases, the factor $(1 - p)^{k-1}$ makes these probabilities decrease geometrically.

Remark on "never seeing a success"

There is a logically possible outcome where every trial is a failure (for example, all tails when tossing a coin), so there is no first success and X would be "infinite". However, the probability of this event is

$$\lim_{k \to \infty} (1 - p)^k = 0,$$

because $0 < 1 - p < 1$. So with probability 1 the random variable X takes a finite value in $\{1, 2, 3, \ldots\}$, and the probabilities $(1 - p)^{k-1} p$ sum to 1.

1.4.7 Expectation

The *expected value* (or expectation, or mean) of a random variable is a single number that describes what value we see "on average" in many independent repetitions of the experiment. For a discrete random variable, it is defined as a weighted average of its possible values, where the weights are the corresponding probabilities.

Definition for a discrete random variable

Let X be a discrete random variable with PMF $p_X(x) = \mathbb{P}(X = x)$. The expected value of X, denoted $\mathbb{E}[X]$, is

$$\mathbb{E}[X] = \sum_x x \, p_X(x),$$

where the sum runs over all possible values of X. Each term is "value $\times$ probability of that value".

If X takes infinitely many (discrete) values, this becomes an infinite series. We then assume that

$$\sum_x |x| \, p_X(x) < \infty,$$

so that $\mathbb{E}[X]$ is well defined and finite.

Example: Game of chance

Suppose a random variable X represents your gain in a single play of a game, with

$$\mathbb{P}(X = 1) = \frac{2}{10}, \quad \mathbb{P}(X = 2) = \frac{5}{10}, \quad \mathbb{P}(X = 4) = \frac{3}{10}.$$

Then

$$\mathbb{E}[X] = 1 \cdot \frac{2}{10} + 2 \cdot \frac{5}{10} + 4 \cdot \frac{3}{10} = \frac{2}{10} + \frac{10}{10} + \frac{12}{10} = \frac{24}{10} = 2.4.$$

If you play many times, your average gain per game will be close to 2.4.

Bernoulli and indicator random variables
Let X be a Bernoulli random variable with parameter p, so

$$\mathbb{P}(X = 1) = p, \qquad \mathbb{P}(X = 0) = 1 - p.$$

Then

$$\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p.$$

As a special case, let $X = I_A$ be the indicator of an event A, so that $X = 1$ if A occurs and $X = 0$ otherwise. Then

$$\mathbb{P}(I_A = 1) = \mathbb{P}(A),$$

and

$$\mathbb{E}[I_A] = \mathbb{P}(A).$$

So the expected value of an indicator random variable equals the probability of the corresponding event.

Uniform random variable on $\{0, 1, \ldots, n\}$
Let X be discrete uniform on $\{0, 1, \ldots, n\}$. Then

$$\mathbb{P}(X = k) = \frac{1}{n+1}, \quad k = 0, 1, \ldots, n.$$

Using the definition,

$$\mathbb{E}[X] = \sum_{k=0}^{n} k \cdot \frac{1}{n+1} = \frac{1}{n+1} \sum_{k=0}^{n} k = \frac{1}{n+1} \cdot \frac{n(n+1)}{2} = \frac{n}{2}.$$

Thus the expectation is the midpoint of the interval $\{0, \ldots, n\}$. More generally, when a PMF is symmetric, the expected value lies at the center of symmetry.

Population-average interpretation
Consider a class of n students, and let x_i be the weight of the i-th student. You perform an experiment where you pick one student uniformly at random, and let X be the weight of the selected student.
The PMF of X is

$$\mathbb{P}(X = x_i) = \frac{1}{n}, \quad i = 1, \ldots, n.$$

Then

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \cdot \frac{1}{n} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

So in this example, the expected value of X is exactly the usual average (arithmetic mean) of the students' weights. This illustrates that expectation can be seen both as the long-run average over many repetitions, and as an average over a finite population when all individuals are equally likely to be chosen.

Elementary properties of expectation
A random variable X can be summarized by its expected value $\mathbb{E}[X]$. Some basic properties of expectation are very natural and are worth stating explicitly.

Nonnegativity
If $X(\omega) \geq 0$ for every outcome ω in the sample space, then

$$\mathbb{E}[X] \geq 0.$$

In words, if the random variable never takes negative values, its average value cannot be negative. This follows because $\mathbb{E}[X]$ is a sum of terms of the form $x\, p_X(x)$, and both factors are nonnegative.

Bounds from the range
Suppose $X(\omega)$ always lies between two constants a and b, that is

$$a \leq X(\omega) \leq b \quad \text{for all } \omega.$$

Then the expectation also lies between a and b:

$$a \leq \mathbb{E}[X] \leq b.$$

To see why $\mathbb{E}[X] \geq a$, write

$$\mathbb{E}[X] = \sum_x x\, p_X(x).$$

Each possible value x of X satisfies $x \geq a$, so

$$\mathbb{E}[X] = \sum_x x\, p_X(x) \geq \sum_x a\, p_X(x) = a \sum_x p_X(x) = a,$$

because the probabilities $p_X(x)$ sum to 1. A symmetric argument with $x \leq b$ gives $\mathbb{E}[X] \leq b$.

Expectation of a constant

A constant c can be viewed as a (degenerate) random variable that always takes the value c. Its PMF assigns probability 1 to the value c and probability 0 to all other values. Then

$$\mathbb{E}[c] = c \cdot 1 = c.$$

So if a quantity is always equal to c, its average is also c.

Expected value rule

A random variable Y is often defined as a function of another random variable X, say

$$Y = g(X).$$

We want a convenient way to compute $\mathbb{E}[Y]$ without first finding the PMF of Y. The *expected value rule* gives exactly this.

Let X be a discrete random variable with PMF $p_X(x) = \mathbb{P}(X = x)$, and let

$$Y = g(X)$$

for some function g. Then

$$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \sum_x g(x)\, p_X(x),$$

where the sum is over all possible values of X.

Conceptually, we sum over the values of X, and for each possible x we take the corresponding value $g(x)$ of Y, weighted by the probability that X equals x.

Example

Suppose X takes values $2, 3, 4, 5$ with probabilities

$$\mathbb{P}(X = 2) = 0.1, \quad \mathbb{P}(X = 3) = 0.2, \quad \mathbb{P}(X = 4) = 0.3, \quad \mathbb{P}(X = 5) = 0.4.$$

Define $Y = g(X)$ by

$$g(2) = 3, \quad g(3) = 3, \quad g(4) = 4, \quad g(5) = 4.$$

Then

$$\mathbb{E}[Y] = \sum_x g(x)\, p_X(x) = 3 \cdot 0.1 + 3 \cdot 0.2 + 4 \cdot 0.3 + 4 \cdot 0.4.$$

This matches the intuitive reasoning: 10

Why the rule works (idea)

By definition,

$$\mathbb{E}[Y] = \sum_y y\, \mathbb{P}(Y = y).$$

For each fixed y, the event $\{Y = y\}$ is the union of all outcomes for which X takes some value x with $g(x) = y$. Thus

$$\mathbb{P}(Y = y) = \sum_{x : g(x) = y} \mathbb{P}(X = x).$$

Substituting this into the definition and rearranging the order of summation leads to

$$\mathbb{E}[Y] = \sum_y y \sum_{x : g(x) = y} \mathbb{P}(X = x) = \sum_x g(x)\, \mathbb{P}(X = x),$$

which is the expected value rule.

A USEFUL SPECIAL CASE
If $g(x) = x^2$, then

$$\mathbb{E}[X^2] = \sum_x x^2 \, p_X(x).$$

In general,

$$\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$$

for most functions g. So, for example, typically

$$\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2.$$

LINEARITY FOR ONE RANDOM VARIABLE
Linearity of expectation describes how expectation behaves under multiplication by constants and addition. It says that taking expectations is a linear operation.
Let X be a random variable and let a, b be constants. Define a new random variable

$$Y = aX + b.$$

Then

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b.$$

DERIVATION
We view Y as a function of X, namely $Y = g(X)$ with $g(x) = ax + b$. By the expected value rule,

$$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \sum_x g(x)\, p_X(x) = \sum_x (ax + b)\, p_X(x).$$

We split the sum:

$$\mathbb{E}[Y] = \sum_x ax\, p_X(x) + \sum_x b\, p_X(x) = a \sum_x x\, p_X(x) + b \sum_x p_X(x).$$

The first sum is $\mathbb{E}[X]$. The second sum is 1, since the probabilities $p_X(x)$ add up to 1. Thus

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b.$$

SPECIAL CASE: AVERAGE SALARY
If X is the salary of a random person in a population, then $\mathbb{E}[X]$ is the average salary. If everyone's salary is updated to

$$Y = 2X + 100,$$

then the new average is

$$\mathbb{E}[Y] = 2\,\mathbb{E}[X] + 100.$$

Doubling all salaries and adding 100 to each one doubles the average and adds 100.

COMMENT
For linear functions $g(x) = ax + b$, we have

$$\mathbb{E}[g(X)] = g(\mathbb{E}[X]).$$

For general, nonlinear functions g, this identity usually does *not* hold.

1.4.8 Variance

Variance measures how much a random variable tends to deviate (in squared distance) from its mean. It complements the expectation by quantifying the spread of the distribution around the average.

DEFINITION
Let X be a random variable with mean $\mu = \mathbb{E}[X]$. The *variance* of X is defined as

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mu)^2\big].$$

This is the expected value of the squared distance from the mean. Since $(X - \mu)^2 \geq 0$ always, the variance is always nonnegative.

Using the expected value rule, for a discrete random variable with PMF $p_X(x)$, we can write

$$\mathrm{Var}(X) = \sum_x (x-\mu)^2 \, p_X(x),$$

where the sum is over all possible values of X. For each value x, we compute its squared distance from μ and weight it by the probability that X takes that value.

Standard deviation

The *standard deviation* of X is defined as

$$\sigma_X = \sqrt{\mathrm{Var}(X)}.$$

It has the same units as X (unlike the variance, which has squared units), and is often easier to interpret as a measure of typical deviation from the mean.

Effect of adding and scaling

Let $Y = aX + b$, where a and b are constants. Then

$$\mathrm{Var}(Y) = a^2 \, \mathrm{Var}(X).$$

In particular,

$$\mathrm{Var}(X+b) = \mathrm{Var}(X), \qquad \mathrm{Var}(aX) = a^2 \, \mathrm{Var}(X)[][].$$

Intuitively, adding a constant b just shifts the distribution right or left without changing its spread, so the variance stays the same. Multiplying by a stretches or shrinks the distribution horizontally by a factor $|a|$, so the spread (variance) scales by a^2

Example

If $\mathrm{Var}(X) = 2$, then

$$\mathrm{Var}(3-4X) = (-4)^2 \, \mathrm{Var}(X) = 16 \cdot 2 = 32.$$

The additive constant 3 does not affect the variance, while the factor -4 multiplies the variance by 16.

Alternative formula

There is a useful identity that often simplifies calculations:

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2[][].$$

It comes from expanding

$$\mathrm{Var}(X) = \mathbb{E}\big[(X-\mu)^2\big] = \mathbb{E}\big[X^2 - 2\mu X + \mu^2\big] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2,$$

since $\mu = \mathbb{E}[X]$. This formula allows us to compute the variance by first finding $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$, and then taking their difference.

Variance of a Bernoulli random variable

A *Bernoulli* random variable and a *discrete uniform* random variable are simple examples where the variance can be computed explicitly. We first recall the Bernoulli case, then the uniform on $\{0, \ldots, n\}$, and finally a general uniform on $\{a, \ldots, b\}$.

Let X be Bernoulli with parameter p, so

$$\mathbb{P}(X=1) = p, \qquad \mathbb{P}(X=0) = 1-p.$$

We know that

$$\mathbb{E}[X] = p.$$

Using the alternative variance formula,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2[][].$$

For a Bernoulli random variable, $X^2 = X$, since $0^2 = 0$ and $1^2 = 1$. Hence

$$\mathbb{E}[X^2] = \mathbb{E}[X] = p,$$

and

$$\mathrm{Var}(X) = p - p^2 = p(1-p).$$

As a function of p, $p(1-p)$ is a parabola on $[0,1]$ with maximum at $p=\frac{1}{2}$ and zeros at $p=0$ and $p=1$. This matches the intuition that a fair coin ($p=\frac{1}{2}$) is "most random", while a deterministic coin ($p=0$ or $p=1$) has variance 0.

Variance of a uniform on $\{0,1,\dots,n\}$
Let X be discrete uniform on $\{0,1,\dots,n\}$. Then

$$\mathbb{P}(X=k)=\frac{1}{n+1},\quad k=0,1,\dots,n.$$

We first compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.
For the mean,

$$\mathbb{E}[X]=\sum_{k=0}^{n}k\cdot\frac{1}{n+1}=\frac{1}{n+1}\sum_{k=0}^{n}k=\frac{1}{n+1}\cdot\frac{n(n+1)}{2}=\frac{n}{2}[][].$$

For $\mathbb{E}[X^2]$,

$$\mathbb{E}[X^2]=\sum_{k=0}^{n}k^2\cdot\frac{1}{n+1}=\frac{1}{n+1}\sum_{k=0}^{n}k^2.$$

The sum of squares satisfies

$$\sum_{k=0}^{n}k^2=\frac{n(n+1)(2n+1)}{6}[][],$$

so

$$\mathbb{E}[X^2]=\frac{1}{n+1}\cdot\frac{n(n+1)(2n+1)}{6}=\frac{n(2n+1)}{6}.$$

Now apply $\mathrm{Var}(X)=\mathbb{E}[X^2]-\big(\mathbb{E}[X]\big)^2$:

$$\mathrm{Var}(X)=\frac{n(2n+1)}{6}-\left(\frac{n}{2}\right)^2=\frac{n(2n+1)}{6}-\frac{n^2}{4}.$$

Bringing to a common denominator,

$$\mathrm{Var}(X)=\frac{4n(2n+1)-3n^2}{12}=\frac{8n^2+4n-3n^2}{12}=\frac{5n^2+4n}{12}.$$

This is the variance of the uniform distribution on $\{0,1,\dots,n\}$ written via the sum-of-squares formula.
In many references, one instead writes the variance of the uniform on $\{0,1,\dots,n\}$ as

$$\mathrm{Var}(X)=\frac{n(n+2)}{12},$$

which is algebraically equivalent after simplifying the sum carefully.

Uniform on $\{a,a+1,\dots,b\}$
Now consider a general discrete uniform random variable Y on $\{a,a+1,\dots,b\}$. Let

$$n=b-a.$$

We can write

$$Y=a+X,$$

where X is uniform on $\{0,1,\dots,n\}$. Adding the constant a does not change the variance, so

$$\mathrm{Var}(Y)=\mathrm{Var}(X)[][].$$

Thus, substituting $n=b-a$,

$$\mathrm{Var}(Y)=\frac{(b-a)\big((b-a)+2\big)}{12},$$

which expresses the variance in terms of the endpoints of the range.

Conditional PMF given an event

Let X be a discrete random variable with PMF $p_X(x) = \mathbb{P}(X = x)$. Let A be an event with $\mathbb{P}(A) > 0$. The *conditional PMF* of X given A is

$$p_{X|A}(x) = \mathbb{P}(X = x \mid A) = \frac{\mathbb{P}(X = x \cap A)}{\mathbb{P}(A)}.$$

This is an ordinary PMF on the same set of values as X, but computed in the conditional model where A is known to have occurred. Its entries satisfy

$$\sum_x p_{X|A}(x) = 1.$$

Conditional expectation given an event

The *conditional expectation* of X given A is the mean of X in the conditional model. It is defined by

$$\mathbb{E}[X \mid A] = \sum_x x\,\mathbb{P}(X = x \mid A) = \sum_x x\,p_{X|A}(x).$$

This is the same formula as for an ordinary expectation, but with conditional probabilities throughout.

Example: Conditioning a uniform

Let X be uniform on $\{1, 2, 3, 4\}$. Then

$$\mathbb{P}(X = x) = \frac{1}{4}, \quad x = 1, 2, 3, 4.$$

By symmetry,

$$\mathbb{E}[X] = \frac{1+4}{2} = \frac{5}{2},$$

and using the uniform-variance formula on $\{1, 2, 3, 4\}$,

$$\mathrm{Var}(X) = \frac{1}{12}(4-1)\big((4-1)+2\big) = \frac{1}{12} \cdot 3 \cdot 5 = \frac{5}{4}.$$

Now condition on the event

$$A = \{X \in \{2, 3, 4\}\}.$$

In the conditional model, outcome 1 is impossible, so

$$\mathbb{P}(X = 1 \mid A) = 0.$$

The remaining values $2, 3, 4$ had equal probabilities in the original model, so they remain equally likely given A. Since their probabilities must sum to 1,

$$\mathbb{P}(X = 2 \mid A) = \mathbb{P}(X = 3 \mid A) = \mathbb{P}(X = 4 \mid A) = \frac{1}{3}.$$

Thus the conditional PMF is uniform on $\{2, 3, 4\}$.
The conditional expectation is the mean of this conditional uniform:

$$\mathbb{E}[X \mid A] = \frac{2+4}{2} = 3.$$

For the conditional variance $\mathrm{Var}(X \mid A)$, we can compute directly. The mean is 3, so

$$\mathrm{Var}(X \mid A) = \mathbb{E}\big[(X-3)^2 \mid A\big] = (2-3)^2 \cdot \frac{1}{3} + (3-3)^2 \cdot \frac{1}{3} + (4-3)^2 \cdot \frac{1}{3} = 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{2}{3}.$$

Notice that

$$\mathrm{Var}(X \mid A) = \frac{2}{3} < \mathrm{Var}(X) = \frac{5}{4}.$$

In the conditional model we have more information (we know that X is not 1), so there is less uncertainty, and the variance decreases.

1.4.9 Total expectation theorem

The Total Expectation Theorem provides a way to compute an expectation by splitting the probability space into scenarios and averaging the conditional expectations over those scenarios. It is the expectation analogue of the Total Probability Theorem.
Let X be a random variable and let A_1, A_2, A_3 be events that form a partition of the sample space, that is:

$$A_i \cap A_j = \emptyset \text{ for } i \neq j, \quad A_1 \cup A_2 \cup A_3 = \Omega, \quad \mathbb{P}(A_i) > 0.$$

Then the *Total Expectation Theorem* says that

$$\mathbb{E}[X] = \mathbb{P}(A_1)\,\mathbb{E}[X \mid A_1] + \mathbb{P}(A_2)\,\mathbb{E}[X \mid A_2] + \mathbb{P}(A_3)\,\mathbb{E}[X \mid A_3].$$

More generally, for a finite or countable partition $\{A_i\}$,

$$\mathbb{E}[X] = \sum_i \mathbb{P}(A_i)\,\mathbb{E}[X \mid A_i].$$

From total probability to total expectation
For a fixed value x, the Total Probability Theorem applied to the event $\{X = x\}$ gives

$$\mathbb{P}(X = x) = \sum_i \mathbb{P}(X = x \mid A_i)\,\mathbb{P}(A_i).$$

This holds for every possible value x. Multiply both sides by x and sum over all x:

$$\sum_x x\,\mathbb{P}(X = x) = \sum_x x \sum_i \mathbb{P}(X = x \mid A_i)\,\mathbb{P}(A_i).$$

The left-hand side is $\mathbb{E}[X]$. On the right-hand side, we can interchange the order of summation:

$$\mathbb{E}[X] = \sum_i \mathbb{P}(A_i) \sum_x x\,\mathbb{P}(X = x \mid A_i).$$

But for each i,

$$\sum_x x\,\mathbb{P}(X = x \mid A_i) = \mathbb{E}[X \mid A_i].$$

Substituting this into the previous expression gives the Total Expectation Theorem.

Interpretation
We can think of $\{A_1, A_2, A_3\}$ as different scenarios. Under each scenario A_i, the random variable X has a conditional distribution and a conditional expectation $\mathbb{E}[X \mid A_i]$. The theorem says that the overall expectation $\mathbb{E}[X]$ is the weighted average of these conditional expectations, where the weights are the scenario probabilities $\mathbb{P}(A_i)$.

Example: Splitting a PMF into two scenarios
Consider a discrete random variable X with PMF having six mass points: three "small" values and three "large" values. Define

$$A_1 = \{\text{“small” values of } X\}, \quad A_2 = \{\text{“large” values of } X\},$$

so that $\{A_1, A_2\}$ is a partition of the sample space.
Suppose the probabilities of the three small values are all $1/9$, so

$$\mathbb{P}(A_1) = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3},$$

and the probabilities of the three large values are all $2/9$, so

$$\mathbb{P}(A_2) = \frac{2}{9} + \frac{2}{9} + \frac{2}{9} = \frac{2}{3}.$$

In the conditional model given A_1, the three small values are equally likely, so X is uniform on $\{0, 1, 2\}$ and

$$\mathbb{E}[X \mid A_1] = 1$$

(the midpoint). Similarly, given A_2, X is uniform on $\{6, 7, 8\}$, so

$$\mathbb{E}[X \mid A_2] = 7.$$

By the Total Expectation Theorem,

$$\mathbb{E}[X] = \mathbb{P}(A_1)\,\mathbb{E}[X \mid A_1] + \mathbb{P}(A_2)\,\mathbb{E}[X \mid A_2] = \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 7 = \frac{1}{3} + \frac{14}{3} = 5.$$

Thus, instead of computing $\mathbb{E}[X]$ directly from all six probabilities at once, we divided the problem into two simpler conditional expectations and combined them with appropriate weights.

1.4.10 Geometric PMF and memorylessness

Let X be the number of tosses until the first heads. Then X takes values $1, 2, 3, \ldots$, and

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \quad k = 1, 2, \ldots$$

The key property is *memorylessness*. Suppose we know that the first toss was tails, so $X > 1$. From that point on, we are again waiting for the first heads in a sequence of independent tosses with success probability p. Thus, the number of *additional* tosses until heads, namely $X - 1$, has the same geometric distribution as X, when we condition on $X > 1$:

$$X - 1 \mid (X > 1) \sim \text{Geometric}(p).$$

More generally, for any integer $n \geq 1$,

$$X - n \mid (X > n) \sim \text{Geometric}(p).$$

Concretely, for example

$$\mathbb{P}(X - 1 = 3 \mid X > 1) = \mathbb{P}(\text{T,T,T,H on tosses } \not\models, \not\models, \not\models \mid \text{T on toss } \not\models).$$

Because tosses are independent, conditioning on tails on the first toss does not change the probabilities of future sequences, so this equals

$$\mathbb{P}(\text{T,T,T,H on tosses } 2, 3, 4) = (1-p)^3 p,$$

which is exactly $\mathbb{P}(X = 4)$, the geometric PMF at $k = 4$. This illustrates that the conditional distribution of the remaining waiting time is geometric with parameter p.

Using memorylessness to find the expectation

Directly computing

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k(1-p)^{k-1}p$$

is possible but involves an infinite series. Instead, we use memorylessness and the total expectation idea.

Write

$$X = 1 + (X - 1).$$

Taking expectations,

$$\mathbb{E}[X] = 1 + \mathbb{E}[X - 1].$$

Now split $\mathbb{E}[X - 1]$ into two scenarios according to the outcome of the first toss:

$$\mathbb{E}[X - 1] = \mathbb{E}[X - 1 \mid X = 1]\,\mathbb{P}(X = 1) + \mathbb{E}[X - 1 \mid X > 1]\,\mathbb{P}(X > 1).$$

If $X = 1$, then $X - 1 = 0$, so

$$\mathbb{E}[X - 1 \mid X = 1] = 0, \quad \mathbb{P}(X = 1) = p.$$

Thus the first term is zero. The second term is

$$\mathbb{E}[X - 1 \mid X > 1]\,\mathbb{P}(X > 1) = \mathbb{E}[X - 1 \mid X > 1](1 - p).$$

By memorylessness,

$$X - 1 \mid (X > 1) \sim \text{Geometric}(p),$$

so

$$\mathbb{E}[X - 1 \mid X > 1] = \mathbb{E}[X].$$

Therefore

$$\mathbb{E}[X - 1] = (1 - p)\,\mathbb{E}[X].$$

Substitute back into $\mathbb{E}[X] = 1 + \mathbb{E}[X - 1]$:

$$\mathbb{E}[X] = 1 + (1 - p)\,\mathbb{E}[X].$$

Rearrange:

$$\begin{aligned} \mathbb{E}[X] - (1-p)\,\mathbb{E}[X] &= 1, \\ p\,\mathbb{E}[X] &= 1, \\ \mathbb{E}[X] &= \frac{1}{p}. \end{aligned}$$

This makes intuitive sense: if the success probability p is small, we expect to wait longer on average until the first heads.

1.4.11 Joint PMFs and the expected value rule

When dealing with several discrete random variables together, their joint behaviour is described by a *joint* PMF. From the joint PMF we can recover the individual (marginal) PMFs and compute expectations of functions of several variables.

Joint and marginal PMFs
Let X and Y be discrete random variables. Their *joint* PMF is

$$p_{X,Y}(x,y) = \mathbb{P}(X = x,\, Y = y).$$

This is a function of two arguments: for each pair (x, y) it gives the probability that X takes value x and Y takes value y simultaneously. It satisfies

$$\sum_x \sum_y p_{X,Y}(x,y) = 1.$$

The *marginal* PMFs of X and Y are obtained by summing out the other variable:

$$p_X(x) = \mathbb{P}(X = x) = \sum_y p_{X,Y}(x,y),$$

$$p_Y(y) = \mathbb{P}(Y = y) = \sum_x p_{X,Y}(x,y).$$

For example, if the joint PMF is given in a table, then $p_X(x)$ is the sum of the entries in the column corresponding to x, and $p_Y(y)$ is the sum of the entries in the row corresponding to y.

Example: reading a joint PMF table
Suppose the joint PMF of X and Y, for $x \in \{0, 1, 2\}$ and $y \in \{1, 2, 3\}$, is given in a 3×3 table of probabilities. Then: - The probability that $X = 1$ and $Y = 3$ is the entry in the row $y = 3$, column $x = 1$. - The marginal $\mathbb{P}(X = 2)$ is the sum of the entries in the column $x = 2$. - The marginal $\mathbb{P}(Y = 2)$ is the sum of the entries in the row $y = 2$.
If we want $\mathbb{P}(X = Y)$, we add the probabilities of all diagonal pairs $(0, 0)$, $(1, 1)$, $(2, 2)$ (or whichever pairs satisfy $x = y$) in the joint table.

Joint PMF for more variables
For three discrete random variables X, Y, Z, the joint PMF is

$$p_{X,Y,Z}(x,y,z) = \mathbb{P}(X = x, Y = y, Z = z),$$

and

$$\sum_x \sum_y \sum_z p_{X,Y,Z}(x,y,z) = 1.$$

From this we can obtain marginals by summing out the variables we are not interested in, for example

$$p_X(x) = \sum_y \sum_z p_{X,Y,Z}(x,y,z),$$

$$p_{X,Y}(x,y) = \sum_z p_{X,Y,Z}(x,y,z).$$

Functions of two random variables and their PMF
Given two random variables X and Y, a function

$$Z = g(X, Y)$$

is itself a random variable. To find the PMF of Z, we use the joint PMF of (X, Y):

$$p_Z(z) = \mathbb{P}(Z = z) = \mathbb{P}\big(g(X,Y) = z\big) = \sum_{(x,y):\, g(x,y)=z} p_{X,Y}(x,y).$$

That is, we sum the probabilities of all pairs (x, y) for which $g(x, y)$ equals the value z.

Expected value rule for functions of two variables
For a function $g(X, Y)$, there is an expected value rule analogous to the one-variable case:

$$\mathbb{E}[g(X,Y)] = \sum_x \sum_y g(x,y)\, p_{X,Y}(x,y).$$

Interpretation: for each possible pair (x, y), the random variables take this pair with probability $p_{X,Y}(x, y)$, and then the function has value $g(x, y)$; the product $g(x, y)\, p_{X,Y}(x, y)$ is the contribution of that pair to the overall average, and we sum over all pairs.

1.4.12 Linearity of expectations

Linearity of expectation extends to sums of multiple random variables and lets us compute the binomial mean very easily.

Linearity for several random variables
For any discrete random variables $X_1, \ldots, X_n$,

$$\mathbb{E}\Big[\sum_{i=1}^{n} X_i\Big] = \sum_{i=1}^{n} \mathbb{E}[X_i].$$

For example, for three variables,

$$\mathbb{E}[2X + 3Y - Z] = 2\,\mathbb{E}[X] + 3\,\mathbb{E}[Y] - \mathbb{E}[Z].$$

Derivation for two variables
For two variables X and Y,

$$\mathbb{E}[X+Y] = \sum_x \sum_y (x+y)\, p_{X,Y}(x,y).$$

Split the sum:

$$\mathbb{E}[X+Y] = \sum_x \sum_y x\, p_{X,Y}(x,y) + \sum_x \sum_y y\, p_{X,Y}(x,y).$$

In the first term, for fixed x,

$$\sum_y x\, p_{X,Y}(x,y) = x \sum_y p_{X,Y}(x,y) = x\, p_X(x),$$

so

$$\sum_x \sum_y x\, p_{X,Y}(x,y) = \sum_x x\, p_X(x) = \mathbb{E}[X].$$

A symmetric argument shows that the second term equals $\mathbb{E}[Y]$. Thus

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

The general n-variable case follows similarly.

Mean of a binomial via indicators
Let $X \sim \text{Binomial}(n,p)$. Interpret X as the number of successes in n independent trials, each with success probability p.
Define indicator variables

$$X_i = \begin{cases} 1 & \text{if trial } i \text{ is a success,} \\ 0 & \text{otherwise,} \end{cases} \qquad i = 1, \ldots, n.$$

Each X_i is Bernoulli(p), so

$$\mathbb{P}(X_i = 1) = p, \quad \mathbb{P}(X_i = 0) = 1 - p, \quad \mathbb{E}[X_i] = p.$$

The total number of successes is the sum of these indicators:

$$X = X_1 + X_2 + \cdots + X_n.$$

By linearity of expectation,

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n] = p + \cdots + p = np.$$

So a binomial random variable with parameters n and p has mean

$$\mathbb{E}[X] = np,$$

which matches the intuition that in n trials with success probability p, we expect np successes on average.

1.4.13 Conditional PMFs

When we condition on the value of another random variable, we obtain a family of conditional PMFs for X, one for each value of Y.

Conditional PMF given another variable
Let X and Y be discrete random variables with joint PMF $p_{X,Y}(x, y)$ and marginal $p_Y(y) = \mathbb{P}(Y = y)$. For any y with $p_Y(y) > 0$, the *conditional PMF* of X given $Y = y$ is

$$p_{X|Y}(x \mid y) = \mathbb{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

If we fix y, this is a PMF in the variable x, and its values satisfy

$$\sum_x p_{X|Y}(x \mid y) = 1.$$

Changing y changes the conditional PMF, so $\{p_{X|Y}(\cdot \mid y)\}$ is a family of PMFs indexed by y.

Example from a joint table
Suppose the joint PMF of (X, Y) is given by a table, and we condition on $Y = 2$. First compute

$$p_Y(2) = \sum_x p_{X,Y}(x, 2),$$

by adding the entries in the row $y = 2$. Then, for each x,

$$p_{X|Y}(x \mid 2) = \frac{p_{X,Y}(x, 2)}{p_Y(2)}.$$

For instance, if the row $y = 2$ has entries proportional to $0, 1, 3, 1$, their sum is 5, so the conditional PMF values are

$$0, \ \frac{1}{5}, \ \frac{3}{5}, \ \frac{1}{5}.$$

Geometrically, the conditional PMF for $X \mid Y = 2$ is the corresponding row of the joint PMF, rescaled so that its entries add to 1.

Multiplication rule in PMF form
From the definition

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)},$$

we can rewrite

$$p_{X,Y}(x, y) = p_Y(y)\, p_{X|Y}(x \mid y),$$

which is the multiplication rule in PMF notation. Symmetrically,

$$p_{X,Y}(x, y) = p_X(x)\, p_{Y|X}(y \mid x).$$

Thus we can specify a joint distribution either directly via $p_{X,Y}$ or indirectly by giving a marginal (say p_Y) and a conditional PMF (say $p_{X|Y}$).

More than two variables
For three random variables X, Y, Z, the conditional PMF of X given $Y = y$ and $Z = z$ is

$$p_{X|Y,Z}(x \mid y, z) = \mathbb{P}(X = x \mid Y = y, Z = z) = \frac{p_{X,Y,Z}(x, y, z)}{p_{Y,Z}(y, z)},$$

where

$$p_{Y,Z}(y, z) = \sum_x p_{X,Y,Z}(x, y, z).$$

We also have a multiplication rule for three variables. Let

$$A = \{X = x\}, \quad B = \{Y = y\}, \quad C = \{Z = z\}.$$

Then

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\, \mathbb{P}(B \mid A)\, \mathbb{P}(C \mid A \cap B),$$

which in PMF notation becomes

$$p_{X,Y,Z}(x, y, z) = p_X(x)\, p_{Y|X}(y \mid x)\, p_{Z|X,Y}(z \mid x, y).$$

CONDITIONAL EXPECTATION
A conditional expectation is just the expectation of X computed in the conditional model "given $Y = y$".
Let X and Y be discrete random variables, and suppose $\mathbb{P}(Y = y) > 0$. The conditional expectation of X given $Y = y$ is

$$\mathbb{E}[X \mid Y = y] = \sum_x x\, \mathbb{P}(X = x \mid Y = y).$$

Here we use the conditional PMF $p_{X|Y}(x \mid y)$ instead of the original PMF of X.

EXPECTED VALUE RULE (CONDITIONAL FORM)
If g is a function and $Z = g(X)$, then in the conditional model "given $Y = y$" we have

$$\mathbb{E}[g(X) \mid Y = y] = \sum_x g(x)\, \mathbb{P}(X = x \mid Y = y).$$

This is the same expected value rule as before, but with conditional probabilities.

TOTAL EXPECTATION THEOREM (VERSION WITH Y)
Let X and Y be discrete random variables, and suppose Y takes values in some countable set $\mathcal{Y}$. Then the total expectation theorem says

$$\mathbb{E}[X] = \sum_{y \in \mathcal{Y}} \mathbb{P}(Y = y)\, \mathbb{E}[X \mid Y = y].$$

So we can view the possible values of Y as "scenarios": for each scenario $Y = y$, we compute the conditional mean $\mathbb{E}[X \mid Y = y]$, and then average these means using $\mathbb{P}(Y = y)$ as weights.

REMARK ON INFINITE RANGES
If Y can take infinitely many discrete values, the sum

$$\sum_y \mathbb{P}(Y = y)\, \mathbb{E}[X \mid Y = y]$$

becomes an infinite series. Under the usual assumption that $\mathbb{E}[X]$ is well defined (finite), this identity still holds, and we can safely use the total expectation theorem even when Y ranges over an infinite discrete set.

1.4.14 Independence of random variables

Independence for random variables extends the idea of independence for events to the level of entire distributions.
Let X be a random variable and A an event. We say that X and A are *independent* if, for every real number x,

$$\mathbb{P}(X = x \text{ and } A) = \mathbb{P}(X = x)\, \mathbb{P}(A).$$

Equivalently, whenever $\mathbb{P}(A) > 0$,

$$\mathbb{P}(X = x \mid A) = \mathbb{P}(X = x) \quad \text{for all } x.$$

Intuition: knowing that A occurred does not change the distribution of X.
Symmetrically, for any x with $\mathbb{P}(X = x) > 0$,

$$\mathbb{P}(A \mid X = x) = \mathbb{P}(A),$$

so learning the value of X does not change your beliefs about A.

INDEPENDENCE OF TWO RANDOM VARIABLES
Let X and Y be discrete random variables. We say that X and Y are *independent* if, for all real x and y,

$$\mathbb{P}(X = x,\, Y = y) = \mathbb{P}(X = x)\, \mathbb{P}(Y = y).$$

In terms of PMFs,

$$p_{X,Y}(x, y) = p_X(x)\, p_Y(y) \quad \text{for all } x, y.$$

So the joint PMF factorizes as the product of the marginals.

Equivalently, for any y with $\mathbb{P}(Y = y) > 0$,

$$\mathbb{P}(X = x \mid Y = y) = \mathbb{P}(X = x) \quad \text{for all } x,$$

that is,

$$p_{X|Y}(x \mid y) = p_X(x).$$

So knowing $Y = y$ does not change the distribution of X. Symmetrically,

$$p_{Y|X}(y \mid x) = p_Y(y),$$

so knowing $X = x$ does not change the distribution of Y.

Independence of multiple random variables

For three random variables X, Y, Z, we say they are *mutually independent* if, for all x, y, z,

$$p_{X,Y,Z}(x, y, z) = p_X(x)\, p_Y(y)\, p_Z(z).$$

More generally, a finite collection $X_1, \ldots, X_n$ is mutually independent if the joint PMF factors as the product of all the marginals:

$$p_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \prod_{i=1}^{n} p_{X_i}(x_i)$$

for all choices of $(x_1, \ldots, x_n)$.

Intuitively, this means that information about some of the variables does not change the distribution of any of the others: all conditional PMFs coincide with the corresponding unconditional PMFs. In practical terms, independence models situations where each random variable is generated by its own separate probabilistic experiment, with no interaction and no shared source of randomness.

Independence gives one more important rule for expectations: products of independent random variables have factorizing expectations.

Product of independent random variables

Let X and Y be independent discrete random variables. Then

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y].$$

Derivation. Using the expected value rule with the joint PMF:

$$\mathbb{E}[XY] = \sum_x \sum_y xy\, p_{X,Y}(x, y).$$

Independence means

$$p_{X,Y}(x, y) = p_X(x)\, p_Y(y),$$

so

$$\mathbb{E}[XY] = \sum_x \sum_y xy\, p_X(x)\, p_Y(y).$$

For fixed x, the factor $x\, p_X(x)$ does not depend on y, hence

$$\mathbb{E}[XY] = \sum_x x\, p_X(x) \sum_y y\, p_Y(y) = \Big(\sum_x x\, p_X(x)\Big)\Big(\sum_y y\, p_Y(y)\Big) = \mathbb{E}[X]\,\mathbb{E}[Y].$$

Products of functions of independent variables

If X and Y are independent, and g and h are functions, then the random variables $g(X)$ and $h(Y)$ are also independent. Therefore,

$$\mathbb{E}[g(X)\, h(Y)] = \mathbb{E}[g(X)]\,\mathbb{E}[h(Y)].$$

We can view this either as a consequence of independence of $g(X)$ and $h(Y)$, or prove it directly by repeating the previous derivation with $g(x)$ in place of x and $h(y)$ in place of y.

We want to understand how variance behaves for sums, and then use this to find the variance of a binomial random variable.

Variance of a sum under independence

In general, $\mathrm{Var}(X+Y)$ is not equal to $\mathrm{Var}(X)+\mathrm{Var}(Y)$. However, if X and Y are independent, we do have

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Assume first that $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, just to simplify the algebra. Then

$$\mathrm{Var}(X+Y) = \mathbb{E}\big[(X+Y)^2\big] = \mathbb{E}\big[X^2 + 2XY + Y^2\big].$$

Using linearity of expectation,

$$\mathrm{Var}(X+Y) = \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2].$$

Since the means are zero, $\mathbb{E}[X^2] = \mathrm{Var}(X)$ and $\mathbb{E}[Y^2] = \mathrm{Var}(Y)$.
If X and Y are independent, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] = 0 \cdot 0 = 0.$$

So

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Without the zero-mean assumption, one can expand $\mathrm{Var}(X+Y)$ using the general formula $\mathrm{Var}(Z) = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2$ and obtain the same result; the cross terms again cancel because $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ under independence.

Examples showing dependence matters

If $Y = X$, then $X + Y = 2X$, so

$$\mathrm{Var}(X+Y) = \mathrm{Var}(2X) = 4\ \mathrm{Var}(X),$$

which is generally not equal to $\mathrm{Var}(X) + \mathrm{Var}(Y) = 2\ \mathrm{Var}(X)$. Here X and Y are completely dependent.
If $Y = -X$, then $X + Y = 0$ always, so

$$\mathrm{Var}(X+Y) = \mathrm{Var}(0) = 0,$$

again very different from $\mathrm{Var}(X) + \mathrm{Var}(Y)$.
These examples show that knowing just the individual variances is not enough; the dependence structure between X and Y is crucial.

Example with independence and scaling

If X and Y are independent, then X is also independent of any function of Y, such as $-3Y$. Thus

$$\mathrm{Var}(X - 3Y) = \mathrm{Var}(X) + \mathrm{Var}(-3Y) = \mathrm{Var}(X) + 9\ \mathrm{Var}(Y),$$

using the rule $\mathrm{Var}(aZ) = a^2\,\mathrm{Var}(Z)$.

Variance of a binomial random variable

Let X be binomial with parameters n and p. Interpret X as the number of successes in n independent trials, each with success probability p.
Define indicator random variables

$$X_i = \begin{cases} 1 & \text{if the } i\text{-th trial is a success,} \\ 0 & \text{otherwise,} \end{cases} \qquad i = 1, \ldots, n.$$

Each X_i is Bernoulli(p), and the total number of successes is

$$X = X_1 + X_2 + \cdots + X_n.$$

The independence of the trials means the indicators $X_1, \ldots, X_n$ are independent.
For each X_i,

$$\mathbb{E}[X_i] = p, \qquad \mathrm{Var}(X_i) = p(1-p).$$

Using independence and the variance-of-sum rule,

$$\mathrm{Var}(X) = \mathrm{Var}\big(X_1 + \cdots + X_n\big) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = \sum_{i=1}^{n} p(1-p) = n\,p(1-p).$$

Thus a binomial random variable with parameters n and p has variance

$$\mathrm{Var}(X) = np(1-p).$$

1.4.15 The hat problem

There are n people and n hats, all permutations equally likely. Let X be the number of people who get their own hat, and X_i the indicator for person i.

Definition of the indicators
For each $i = 1, \ldots, n$,

$$X_i = \begin{cases} 1 & \text{if person } i \text{ gets their own hat,} \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$X = X_1 + X_2 + \cdots + X_n.$$

Expectation of each X_i
In a random permutation, person i is equally likely to receive any of the n hats. Only one of these is their own hat, so

$$\mathbb{P}(X_i = 1) = \frac{1}{n}.$$

Thus

$$\mathbb{E}[X_i] = 1 \cdot \frac{1}{n} + 0 \cdot \left(1 - \frac{1}{n}\right) = \frac{1}{n}.$$

Expectation of X
By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{1}{n} = 1.$$

So in the hat problem, the expected number of people who get their own hat back is always

$$\mathbb{E}[X] = 1,$$

independent of n.

1.5 Continuous random variables

A continuous random variable takes values over a continuous range, such as the real line or an interval on the real line. In the discrete case, one unit of probability is distributed over the possible values as point masses, and the PMF tells us how much probability sits on top of each value. In the continuous case, one unit of probability is still assigned to the possible values, but now it is spread continuously over the real line.
Some parts of the real line receive more probability mass per unit length, others receive less. The *probability density function* (PDF), denoted $f_X(x)$, describes how densely probability is distributed along the real line. The letter f indicates we are dealing with a PDF, and the subscript indicates which random variable we are considering.

Interpretation of the PDF
Think of probability as snowfall. One pound of snow has fallen on the real line, and the PDF tells us the height of the snow accumulated over each point. The weight of snow sitting on top of an interval $[a, b]$ is found by calculating the area under the PDF curve over that interval. Mathematically, area under the curve is an integral, so

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\, dx.$$

This is the probability that X takes values in the interval $[a, b]$.

Properties of a PDF
By analogy with the discrete case, a PDF must satisfy two conditions:

$$f_X(x) \geq 0 \quad \text{for all } x,$$

$$\int_{-\infty}^{+\infty} f_X(x)\, dx = 1.$$

The first condition ensures we do not obtain negative probabilities. The second condition ensures that the total probability over the entire real line equals one, since X is certain to lie somewhere on the real line. These two conditions are all that we need for a legitimate PDF.

Formal definition of a continuous random variable
A continuous random variable is a random variable whose probabilities can be described by a PDF. That is, there exists a function f_X such that

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\, dx$$

for any interval $[a, b]$. It is not enough for a random variable to take values on a continuous set. A continuous random variable must also be describable by a PDF according to this formula.

Probabilities of more complicated sets
Once we have the probabilities of intervals, we can use additivity to calculate the probabilities of more complicated sets. For example, suppose we want the probability that X lies between 1 and 3 or between 4 and 5. This is the probability of a region consisting of two disjoint intervals. By additivity,

$$\mathbb{P}(1 \leq X \leq 3 \text{ or } 4 \leq X \leq 5) = \mathbb{P}(1 \leq X \leq 3) + \mathbb{P}(4 \leq X \leq 5).$$

Each of the probabilities on the right is calculated using the PDF:

$$= \int_1^3 f_X(x)\, dx + \int_4^5 f_X(x)\, dx.$$

The role of the sample space
There is still an underlying sample space lurking in the background. Different outcomes in the sample space result in different numerical values for the random variable X. When we talk about the probability that X takes values in an interval $[a, b]$, we mean the probability of those outcomes for which the resulting value of X lies in that interval.
However, once we have a PDF in our hands, we can completely forget about the sample space and carry out any calculations just by working with the PDF. As we move on in this course, the sample space will appear less and less, and we will work directly with PDFs or PMFs.

Probabilities of small intervals
Let us look at an interval that starts at some point a and goes up to $a+\delta$, where δ is a small positive number. The probability that X falls in this interval is the area under the PDF over the interval $[a, a+\delta]$. As long as f_X does not change too much over this little interval, which will be the case if f_X is continuous, we can approximate the area by the area of a rectangle with constant height. The area of this rectangle is

$$\mathbb{P}(a \leq X \leq a+\delta) \approx f_X(a) \cdot \delta.$$

This gives us an interpretation of PDFs in terms of probabilities of small intervals.
If we rearrange this approximation, we see that

$$f_X(a) \approx \frac{\mathbb{P}(a \leq X \leq a+\delta)}{\delta}.$$

So the value of the PDF can be interpreted as *probability per unit length.* PDFs are not probabilities, they are densities, and their units are probability per unit length.

Probability of a single point
If the probability per unit length is finite and the length δ is sent to zero, we obtain zero probability. More formally, if we let $b = a$ in the integral

$$\mathbb{P}(a \leq X \leq a) = \int_a^a f_X(x)\, dx,$$

we get an integral over a zero-length interval, and that integral is zero. Therefore,

$$\mathbb{P}(X = a) = 0$$

for any particular point a.
So for a continuous random variable, any particular point has zero probability. Yet, collectively, the infinitely many points in an interval together have positive probability. This is not a puzzle, it is exactly what happens with the ordinary notion of length. Single points have zero length, yet by putting together lots of points we can create a set with positive length.

Open and closed intervals
A final consequence of the fact that individual points have zero probability is that it does not matter whether we include the endpoints of an interval or not. Using additivity,

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}(X = a) + \mathbb{P}(a < X < b) + \mathbb{P}(X = b).$$

Since $\mathbb{P}(X = a) = 0$ and $\mathbb{P}(X = b) = 0$, we conclude that

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}(a < X < b).$$

When calculating probabilities of intervals, it does not matter whether we include the endpoints or not.

1.5.1 Uniform and piecewise constant PDFs

The uniform random variable is a basic example of a continuous random variable. In the discrete uniform case, the possible values are integers between a and b, all equally likely. In the continuous case, any real number between a and b is possible, and the PDF has the same height at all points in this interval.
Intuitively, a uniform random variable models a situation where we know that the value lies between a and b, but we have no reason to believe that any location inside the interval is more likely than another. In this sense, it represents a situation of complete ignorance.

PDF of a uniform random variable
Let X be uniform on $[a, b]$. Its PDF is constant on $[a, b]$ and zero elsewhere:

$$f_X(x) = \begin{cases} c, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

The total probability must be 1, so the area under the PDF is

$$1 = \int_a^b c\, dx = c(b-a),$$

which gives $c = \frac{1}{b-a}$. Thus

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

Any two intervals of the same length inside $[a, b]$ have the same probability, because their probabilities are the base (length) times the same constant height.

Piecewise constant PDFs
A more general kind of PDF is piecewise constant. In this case, the PDF is constant on each of several subintervals, possibly with different heights on different intervals. This shows that PDFs need not be continuous; they may have jumps and discontinuities.
For such a PDF to be valid, the total area under the curve must be 1. This total area is the sum of the areas of the rectangles corresponding to each constant piece.

Computing probabilities
With a piecewise constant PDF, probabilities are obtained by adding areas of rectangles. If a certain event corresponds to an interval that intersects several constant pieces, its probability is the sum of the areas over each part of that interval. For example, if over two subintervals the heights are h_1 and h_2 and the lengths are ℓ_1 and ℓ_2, then the probability of their union is

$$h_1\ell_1 + h_2\ell_2.$$

1.5.2 Means and variances

We now develop continuous counterparts of the concepts of expectation and variance. In the discrete case, expectation is a weighted average of the possible values of a random variable, weighted by their probabilities. In the continuous case, we use the same idea, but the sum is replaced by an integral and the weights come from the PDF.

Expectation of a continuous random variable
Let X be a continuous random variable with PDF $f_X(x)$. Its expectation (or mean) is defined by

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x)\, dx.$$

This is a weighted average of the possible values of X, where points with higher density $f_X(x)$ contribute more. More generally, for any function g,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x)\, dx.$$

This is the continuous analogue of the discrete expected value rule.
To ensure that the expectation is well defined, we usually assume that

$$\int_{-\infty}^{+\infty} |x| f_X(x)\, dx < \infty.$$

Under this assumption, $\mathbb{E}[X]$ exists.
The intuition of expectation remains the same as in the discrete case. It represents the average value we would see if we repeated the experiment many times independently. It can also be viewed as the center of gravity of the distribution. If the PDF is symmetric around a point c, then the expectation is c.

Basic properties of expectation
Expectations of continuous random variables satisfy the same properties as in the discrete case. If $X \ge 0$ always, then $\mathbb{E}[X] \ge 0$. If X always lies in an interval $[a, b]$, then $\mathbb{E}[X]$ also lies in $[a, b]$.
A key property is linearity of expectation. For constants a and b,

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b.$$

Linearity holds regardless of whether X is discrete or continuous.

Variance and standard deviation
The variance of a continuous random variable is defined exactly as in the discrete case:

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mu)^2\big],$$

where $\mu = \mathbb{E}[X]$. Using the expected value rule, this becomes

$$\mathrm{Var}(X) = \int_{-\infty}^{+\infty} (x-\mu)^2 f_X(x)\, dx.$$

The standard deviation is

$$\sigma = \sqrt{\mathrm{Var}(X)}.$$

Variance measures how spread out the values of X are around the mean. A small variance means that most of the probability mass is concentrated close to μ. A large variance means that values far from μ occur more often.

Alternative formula for the variance
It is often convenient to use the formula

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2.$$

Here

$$\mathbb{E}[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x)\, dx.$$

This alternative expression is derived exactly as in the discrete case and is usually easier to use in calculations.

Effect of linear transformations
If we transform a random variable X by a linear function $aX + b$, the variance behaves as follows:

$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X).$$

Adding a constant b does not change the variance. Multiplying by a constant a scales the variance by a^2.

1.5.3 Mean and variance of the uniform

We now illustrate mean and variance calculations using the continuous uniform random variable. This is the continuous analogue of the discrete uniform, for which we already know formulas for the mean and variance.

Mean of the uniform
Let X be uniform on $[a, b]$, with density

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

The mean is

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x)\, dx.$$

Since $f_X(x) = 0$ outside $[a, b]$, this reduces to

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a}\, dx = \frac{1}{b-a} \int_a^b x\, dx.$$

We have

$$\int_a^b x\, dx = \frac{b^2}{2} - \frac{a^2}{2},$$

so

$$\mathbb{E}[X] = \frac{1}{b-a}\left(\frac{b^2}{2} - \frac{a^2}{2}\right) = \frac{a+b}{2}.$$

This is the same as in the discrete case and also matches the midpoint of the interval $[a, b]$, reflecting the symmetry of the PDF around $\frac{a+b}{2}$.

Computing $\mathbb{E}[X^2]$
To use the alternative variance formula, we first compute $\mathbb{E}[X^2]$. By the expected value rule,

$$\mathbb{E}[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x)\, dx = \int_a^b x^2 \cdot \frac{1}{b-a}\, dx.$$

Thus

$$\mathbb{E}[X^2] = \frac{1}{b-a}\int_a^b x^2\,dx.$$

Since

$$\int_a^b x^2\,dx = \frac{b^3}{3} - \frac{a^3}{3},$$

we obtain

$$\mathbb{E}[X^2] = \frac{1}{b-a}\left(\frac{b^3}{3} - \frac{a^3}{3}\right).$$

The cubic terms appear because the antiderivative of x^2 is $\frac{x^3}{3}$.

Variance and standard deviation

Using the alternative formula

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2,$$

we substitute

$$\mathbb{E}[X^2] = \frac{1}{b-a}\left(\frac{b^3}{3} - \frac{a^3}{3}\right), \quad \mathbb{E}[X] = \frac{a+b}{2}.$$

After some algebra, this simplifies to

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

The standard deviation is then

$$\sigma = \sqrt{\mathrm{Var}(X)} = \frac{b-a}{\sqrt{12}}.$$

A useful observation is that the standard deviation is proportional to the width $b-a$ of the interval. The wider the uniform distribution, the larger the standard deviation, which matches the intuition that standard deviation measures the spread or width of the distribution. The variance grows with the square of the length $b-a$, reflecting that doubling the width multiplies the variance by four.

1.5.4 Exponential random variables

We now introduce a new continuous random variable, the exponential random variable. It is determined by a single positive parameter $\lambda > 0$, and its PDF is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Since the PDF is zero for negative x, the exponential random variable is always non-negative.
For $x \geq 0$, the PDF starts at value λ when $x = 0$ and then decays exponentially. When λ is small, the initial height is small but the decay is slow, so the distribution extends over a wide range. When λ is large, the PDF starts high near zero and decays quickly, so most of the probability is concentrated near 0.
The shape of the exponential PDF is analogous to that of the geometric distribution in the discrete case. There, probabilities decrease geometrically with the value; here they decrease exponentially.

Tail probabilities

Fix $a > 0$. We want to compute

$$\mathbb{P}(X \geq a) = \int_a^\infty \lambda e^{-\lambda x}\,dx.$$

Using the antiderivative $\int e^{-\lambda x}dx = -\frac{1}{\lambda}e^{-\lambda x}$, we get

$$\mathbb{P}(X \geq a) = \lambda\left[-\frac{1}{\lambda}e^{-\lambda x}\right]_{x=a}^{x=\infty} = -e^{-\lambda x}\Big|_{x=a}^{x=\infty} = 0 - \big(-e^{-\lambda a}\big) = e^{-\lambda a}.$$

Thus the tail probability decays exponentially in a.
If we set $a = 0$, we obtain

$$\mathbb{P}(X \geq 0) = e^{-\lambda \cdot 0} = 1,$$

which confirms that the total integral of the PDF over its support is 1.

Mean and second moment

To compute the mean, we use the definition

$$\mathbb{E}[X] = \int_0^\infty x\lambda e^{-\lambda x}\, dx.$$

This integral can be evaluated by integration by parts and the result is

$$\mathbb{E}[X] = \frac{1}{\lambda}.$$

For the second moment, we similarly write

$$\mathbb{E}[X^2] = \int_0^\infty x^2\lambda e^{-\lambda x}\, dx,$$

which is again computed by integration by parts (a bit more tedious). The result is

$$\mathbb{E}[X^2] = \frac{2}{\lambda^2}.$$

Variance
Using the alternative variance formula,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2,$$

we substitute

$$\mathbb{E}[X^2] = \frac{2}{\lambda^2}, \quad \mathbb{E}[X] = \frac{1}{\lambda}.$$

Thus

$$\mathrm{Var}(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

When λ is small, the PDF decays slowly, so large values of X are more likely. In that case, both the mean $\frac{1}{\lambda}$ and the variance $\frac{1}{\lambda^2}$ are large, reflecting a wide and spread-out distribution.

Interpretation and relation to geometric
The exponential distribution is closely related to the geometric distribution. The mean $\mathbb{E}[X] = 1/\lambda$ parallels the geometric mean $1/p$, where p is the success probability. The exponential is often used to model waiting times until an event occurs, such as the time until a customer arrives, a machine fails, a light bulb burns out, or an email arrives.

1.5.5 Cumulative distribution functions

We now introduce a way to describe the distribution of any random variable, discrete or continuous, using the *cumulative distribution function* (CDF). For a random variable X, the CDF is the function

$$F_X(x) = \mathbb{P}(X \le x).$$

It gives the probability that X takes a value less than or equal to x.
The CDF is defined the same way for all types of random variables. If X is continuous with PDF f_X, then

$$F_X(x) = \int_{-\infty}^x f_X(t)\, dt.$$

Example: uniform random variable
Let X be uniform on $[a, b]$. The CDF $F_X(x)$ has three cases:

$$F_X(x) = \begin{cases} 0, & x \le a, \\ \dfrac{x-a}{b-a}, & a < x < b, \\ 1, & x \ge b. \end{cases}$$

For $x \le a$, there is no probability mass to the left, so the CDF is 0. For $a < x < b$, the probability up to x is the area of a rectangle with base $x - a$ and height $\frac{1}{b-a}$, giving $\frac{x-a}{b-a}$. For $x \ge b$, we have already covered all the probability, so the CDF is 1.
The graph starts at 0, rises linearly from a to b, and then stays at 1.

Using the CDF to get probabilities
Once we know the CDF, we can compute probabilities of intervals. For example,

$$\mathbb{P}(3 < X \leq 4) = F_X(4) - F_X(3).$$

More generally, for any $a < b$,

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

Thus the CDF contains all probabilistic information about the distribution.
For continuous random variables, we can recover the PDF from the CDF wherever the derivative exists:

$$f_X(x) = \frac{d}{dx} F_X(x).$$

Conversely, given the PDF, we obtain the CDF by integrating.

CDFs for discrete random variables
For a discrete random variable X, the CDF is still

$$F_X(x) = \mathbb{P}(X \leq x),$$

but it is computed as a sum of PMF values for all outcomes k with $k \leq x$. The CDF has the shape of a staircase: it is constant between the possible values of X and jumps upward at each value where X has positive probability.
At a point x_0 where X can take the value x_0, the jump size is

$$F_X(x_0) - \lim_{x \uparrow x_0} F_X(x) = \mathbb{P}(X = x_0).$$

The value of the CDF at a jump is the value *including* that point, because the definition uses $\leq x$.

General properties of CDFs
CDFs satisfy some general properties for any random variable:

$$\text{(i) } F_X(x) \text{ is non-decreasing in } x,$$

$$\text{(ii) } \lim_{x \to -\infty} F_X(x) = 0,$$

$$\text{(iii) } \lim_{x \to +\infty} F_X(x) = 1.$$

The non-decreasing property reflects that larger x correspond to larger events $\{X \leq x\}$. As we move far to the left, the probability of being $\leq x$ must go to 0, and as we move far to the right, the probability must approach 1.

1.5.6 Normal random variables

Normal random variables, also called Gaussian random variables, are among the most important in probability theory. They play a key role in theoretical results such as the central limit theorem and are widely used in applications as models of random noise.

Standard normal distribution
The *standard normal* random variable is denoted by

$$X \sim \mathcal{N}(0, 1).$$

Its PDF is defined for all real x by

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty.$$

This random variable can take any real value.
The factor $e^{-x^2/2}$ has a bell shape centered at 0: it is equal to 1 at $x = 0$ and decreases as $|x|$ increases. The constant $\frac{1}{\sqrt{2\pi}}$ is chosen so that the total area under the curve is 1.
Because the PDF is symmetric around 0, the mean of the standard normal is

$$\mathbb{E}[X] = 0.$$

A calculus computation shows that its variance is

$$\mathrm{Var}(X) = 1.$$

General normal distributions
A general normal random variable with mean μ and variance σ^2 is denoted by

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

where $\sigma > 0$. Its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

The exponent $-\frac{(x-\mu)^2}{2\sigma^2}$ is a quadratic centered at $x = \mu$, so the PDF has a bell shape centered at μ. By symmetry, the mean is

$$\mathbb{E}[X] = \mu.$$

A calculus exercise shows that the variance is

$$\mathrm{Var}(X) = \sigma^2.$$

The parameter σ controls the spread. When σ is small, the quadratic in the exponent rises quickly, the bell is narrow, and the variance is small. When σ is large, the bell is wide and the variance is large.

Linear transformations of normal variables
Normal random variables behave very nicely under linear transformations. Let

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

and define

$$Y = aX + b$$

for constants a and b.
From linearity of expectation,

$$\mathbb{E}[Y] = a\,\mathbb{E}[X] + b = a\mu + b.$$

From the variance rule for linear functions,

$$\mathrm{Var}(Y) = a^2\mathrm{Var}(X) = a^2\sigma^2.$$

The key additional fact is that Y is itself normal:

$$Y \sim \mathcal{N}(a\mu + b,\, a^2\sigma^2).$$

There is one special case. If $a = 0$, then $Y = b$ is a constant (a degenerate discrete random variable with variance 0). By convention, we can view this as a degenerate normal with mean b and variance 0, so that any linear function of a normal random variable is still considered normal.

1.5.7 Calculation of normal probabilities

Normal random variables are very important, so we would like to calculate probabilities such as $\mathbb{P}(X \leq 5)$ when X is normal. In general, the CDF of a normal random variable does not have a closed form expression in terms of elementary functions.

Standard normal CDF and its table
For the standard normal random variable $Y \sim \mathcal{N}(0, 1)$, we denote the CDF by

$$\phi(y) = \mathbb{P}(Y \leq y).$$

Graphically, $\phi(y)$ is the area under the standard normal PDF to the left of y. Values of $\phi(y)$ are tabulated in standard normal tables.
To use such a table, we locate the row and column corresponding to the first two decimal digits of y. For example, $\phi(0) = 0.5$ by symmetry of the standard normal around 0. If we want $\phi(1.16)$, we find the row for 1.1 and the column for 0.06, and read off the corresponding entry. For $y = 2.9$, we find the row for 2.9 and the column for 0.00, which gives $\phi(2.9)$.
Typical tables list $\phi(y)$ only for $y \geq 0$. To handle negative arguments, we use symmetry of the standard normal:

$$\phi(-y) = \mathbb{P}(Y \leq -y) = 1 - \mathbb{P}(Y \leq y) = 1 - \phi(y), \quad y \geq 0.$$

Thus, to find $\phi(-2)$, we compute $1 - \phi(2)$.

Standardization of a general normal

Let X be a normal random variable with mean μ and standard deviation $\sigma > 0$, that is $X \sim \mathcal{N}(\mu, \sigma^2)$. We define the standardized random variable

$$Y = \frac{X - \mu}{\sigma}.$$

The quantity Y measures how many standard deviations X is away from its mean.
The mean of Y is

$$\mathbb{E}[Y] = \frac{\mathbb{E}[X] - \mu}{\sigma} = 0,$$

and the variance of Y is

$$\mathrm{Var}(Y) = \frac{1}{\sigma^2}\mathrm{Var}(X) = 1.$$

If X is normal, then Y is a standard normal random variable. Equivalently,

$$X = \mu + \sigma Y, \quad Y \sim \mathcal{N}(0, 1).$$

Using standardization to compute probabilities

To compute probabilities for X, we transform the event into one involving Y. For example, for any real a and b with $a < b$,

$$\mathbb{P}(a \le X \le b) = \mathbb{P}(a \le \mu + \sigma Y \le b) = \mathbb{P}\left(\frac{a - \mu}{\sigma} \le Y \le \frac{b - \mu}{\sigma}\right).$$

If we write

$$\alpha = \frac{a - \mu}{\sigma}, \quad \beta = \frac{b - \mu}{\sigma},$$

then

$$\mathbb{P}(a \le X \le b) = \mathbb{P}(\alpha \le Y \le \beta) = \phi(\beta) - \phi(\alpha).$$

We can then use the standard normal table to find $\phi(\beta)$ and $\phi(\alpha)$, handling negative arguments via symmetry if needed.

Example of the method

Suppose $X \sim \mathcal{N}(6, 4)$, so $\mu = 6$ and $\sigma = 2$. We want

$$\mathbb{P}(2 \le X \le 8).$$

We standardize:

$$\frac{2 - 6}{2} = -2, \quad \frac{8 - 6}{2} = 1.$$

So

$$\mathbb{P}(2 \le X \le 8) = \mathbb{P}(-2 \le Y \le 1),$$

where $Y \sim \mathcal{N}(0, 1)$. This equals

$$\mathbb{P}(Y \le 1) - \mathbb{P}(Y \le -2) = \phi(1) - \phi(-2).$$

Using symmetry,

$$\phi(-2) = 1 - \phi(2),$$

so

$$\mathbb{P}(2 \le X \le 8) = \phi(1) - \big(1 - \phi(2)\big) = \phi(1) + \phi(2) - 1.$$

Finally, we read $\phi(1)$ and $\phi(2)$ from the standard normal table and substitute the corresponding numerical values.